

UNIT 1



In this unit, we discuss forces and effect of forces on particle and rigid bodies in detail. Particularly, we learn how to determine resultant force of different force systems. The moment of force and couple and concept of equivalent system are also introduced in this unit. Systematically understanding of these concepts is important to the design engineers.

Co-planer System of Forces

1.1 INTRODUCTION TO FORCE

Study of Statics and the whole study of Mechanics is actually the study about the actions of forces or force systems and the effect of these actions on bodies. So it is important to understand the action of forces, characteristics of force systems, and particular methods to analyze them.

What is force?

Force is the action of a body about another body, and it changes or tends to change the state of rest or motion of a body. Forces exist because of an interaction of one object with another object. Whenever there is an interaction between two objects, there is a force upon each of the objects.

Force is an action that has the ability to change motion

Force is measured using standard metric unit known as the Newton; Newton is abbreviated as 'N', so 15 N means 15 Newton of force. One Newton equals 1 kilogram multiplied by 1 meter per second squared. This means that a force of one Newton causes a 1-kilogram mass to have an acceleration of 1 m/sec².

$$1 \text{ N} = (1 \text{ kg}) (1 \text{ m/s}^2) = 1 \text{ kg-m/s}^2.$$

The force is a vector quantity as its effect depends on the direction as well as on the magnitude of the action. Thus, for example 15 N is not a full description of the force acting on an object, in contrast 15 N acting in downward direction (or any direction) is a complete description of the force acting on an object.

Force is a vector quantity

Force acting on an object may cause the object to change its shape, to start moving, to stop moving, to accelerate or decelerate.

As shown in *figure 1.1*, the effect of the force applied to the bracket depends on P , the angle θ and the location of the point of application. If any of these specifications will change, it alters the effect on the bracket. So magnitude, direction, sense and point of application are characteristics of the force.

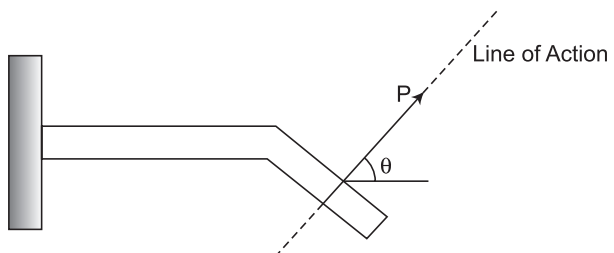


Fig. 1.1

Magnitude represents the value of force, like, 15 N, 10 KN etc. It is also expressed in Pounds (lbs). The Magnitude can be represented graphically by drawing a vector to scale.

1.4 Engineering Mechanics Statics

Direction of the force is represented by line of action and angle it forms with some reference axis. The line of action is indefinitely long line on which the force is laying.

Sense of the force is represented by an arrowhead; it specifies the direction in which the force moves along the line of action. The direction relates to the line of action of the force, and the sense is the way in which the force moves along that line.

Point of application is the exact contact point (location) at which a force is applied to a body. Graphically it is represented by the tip of the arrowhead (it may be situated in the opposite end as the arrowhead) and it is unique to each force. It may happen that different forces share same point of application.

1.2 CLASSIFICATION OF FORCE

Forces are classified as either contact or body forces. Contact force is produced by direct physical contact of two objects, and they are distributed over a surface area of the body; examples of contact forces include applied force, frictional forces, normal forces etc. Body force is produced when body (object) is located in a force field such as a gravitational, electric or magnetic field. These types of forces results when the two interacting objects are not in physical contact with each other and they are distributed over the volume of the body.

Forces are also considered as an external force or internal force. External force is applied externally to an object; they are either applied or reactive forces. Internal force is developed inside the body to resist deformation of body.

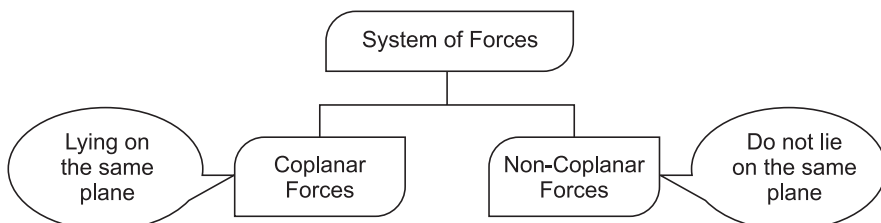
Forces may be further classified as concentrated or distributed force. When the area over which contact force is applied is very small like a point, the force may be considered as concentrated on a point. The force which is distributed over an area is considered as distributed force.

1.2.1 System of Forces

When two or more forces act on a body they are called to form a system of forces. Most practical problems of statics involve system of force.

System of forces is simply a term used to describe a group of forces. System of forces may be classified as follows,

System of forces is simply a particular set of forces

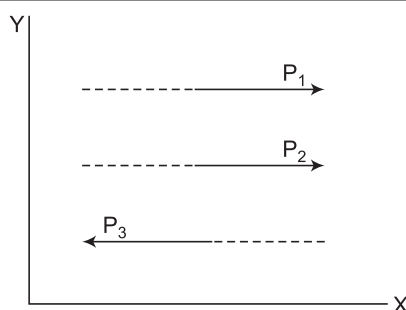


Coplanar forces are those forces whose lines of action lies on the same plane. Coplanar forces may be collinear, concurrent, non-concurrent, and parallel. Non-Coplanar forces are those forces whose lines of action does not lie on the same plane. Non-Coplanar forces may be concurrent, non-concurrent, and parallel. Non-coplanar forces are also known as space forces or spatial force system, as forces passes through space when they change their plane. *Table 1.1* shows types and characteristics of different system of forces.

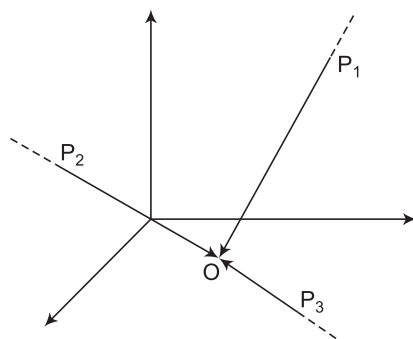
Table 1.1. *Characteristics of Different System of Forces*

Collinear Forces are those forces whose lines of action lies on the same line. As shown in figure, forces P_1 and P_2 acting in same plane and having a common line of action, this system is coplanar collinear force system. Similarly forces P_3 and P_4 are collinear forces.	
Coplanar Concurrent Forces are those forces whose lines of action passes through the same point and also lie on the same plane. The concurrent forces may or may not be collinear. As shown in figure, line of action of forces P_1, P_2 and P_3 meet at common point O and also these forces lie on the same plane, this system is coplanar concurrent force system.	
Coplanar Non-Concurrent Forces are those forces whose lines of action does not pass through the same point but lie on the same plane. As shown in figure, forces P_1, P_2, P_3 and P_4 lie in the same plane but their lines of action do not meet at same point, this system is coplanar non-concurrent force system.	

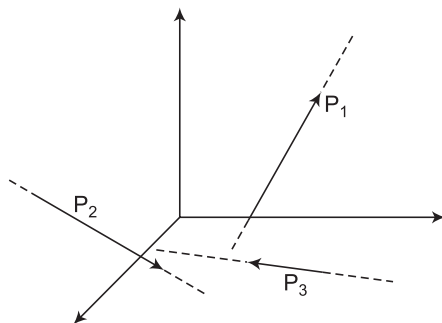
Coplanar Parallel Forces are those forces whose lines of action are parallel to each other and also lie on the same plane. As shown in figure, forces P_1 , P_2 , and P_3 lie in the same plane and their lines of action are parallel to each other, this system is coplanar parallel force system. Parallel forces can be in the same or in opposite directions. Parallel forces which are having same direction are known as like parallel forces and parallel forces which are having different directions are known as unlike parallel forces.



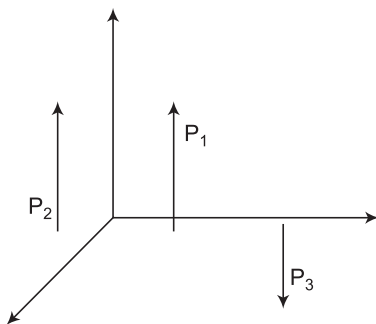
Non-Coplanar Concurrent Forces are those forces whose lines of action pass through the same point but do not lie on the same plane. As shown in figure, line of action of forces P_1 , P_2 and P_3 meet at common point O but do not lie on the same plane, this system is non-coplanar concurrent force system.



Non-Coplanar Non-Concurrent Forces are those forces whose lines of action do not pass through the same point and also not lie on the same plane. As shown in figure, line of action of forces P_1 , P_2 and P_3 do not pass through the same point and also not lie on the same plane, this system is non-coplanar non-concurrent force system.



Non-Coplanar Parallel Forces are those forces whose lines of action are parallel to each other but do not lie in the same plane. As shown in figure, forces P_1 , P_2 , and P_3 are parallel to each other but do not lie in the same plane.



1.3 RESULTANT FORCE

When number of forces are acting on an object then it is possible to replace them with a single force which produces the same effect as produced by all the given forces, this single force is known as

Composition is the procedure to find out single resultant force of a system of forces

resultant force.

Resultant force is denoted by R . The procedure to find out single resultant force of a system of forces is known as composition of forces.

Resultant is a single force which produces the same effect as produced by number of forces when acting together

1.3.1 Resultant of Collinear Forces

The resultant of two or more collinear forces is simply their algebraic sum and acts on the same line on which lines of action of all collinear forces lie. If two forces P_1 and P_2 act in same direction as shown in figure 1.2, their resultant $R = P_1 + P_2$. If two forces P_1 and P_2 act in opposite direction as shown in figure 1.3, their resultant $R = P_1 - P_2$. As a matter of convenience, force acting towards right is considered as positive and force acting toward left is considered as negative.



Fig. 1.2



Fig. 1.3

1.3.2 Resultant of Concurrent Forces

Here we may consider two cases of concurrent forces. Case I, resultant of two concurrent forces and Case II, resultant of more than two concurrent forces.

CASE I. Resultant of two concurrent forces:

Resultant of two concurrent forces can be found out by means of the parallelogram law or the triangle law. These methods are discussed in following section:

Parallelogram Law of Forces (Analytical method)

It state, "If two forces acting simultaneously at a point be represented in magnitude and direction by two adjacent sides of a parallelogram, their resultant may be represented in magnitude and direction by the diagonal of the parallelogram passing through that point."

In figure 1.4, two forces P_1 and P_2 acting at point O are represented

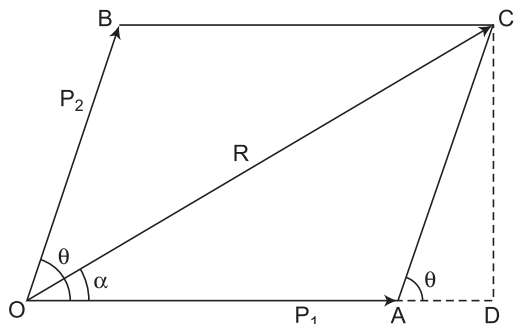


Fig. 1.4

in magnitude and direction by OA and OB respectively. By parallelogram law the resultant R is represented in magnitude and direction by OC .

We now find magnitude and direction of resultant as below:

Let, θ be the angle between two forces P_1 and P_2 and α is the angle between force P_1 and resultant R .

Now from C draw a perpendicular CD to OA produced.

Now $\angle DAC = \angle AOB = \theta$... (Corresponding angle)

In $\triangle DAC$, $CD = P_2 \sin \theta$ and $AD = P_2 \cos \theta$

In $\triangle DOC$,

$$OC^2 = OD^2 + CD^2 = (OA + AD)^2 + CD^2$$

$$\therefore R^2 = (P_1 + P_2 \cos \theta)^2 + (P_2 \sin \theta)^2$$

$$\therefore R^2 = P_1^2 + 2P_1P_2 \cos \theta + P_2^2 \cos^2 \theta + P_2^2 \sin^2 \theta$$

$$\therefore R^2 = P_1^2 + 2P_1P_2 \cos \theta + P_2^2 (\cos^2 \theta + \sin^2 \theta)$$

$$\therefore R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta} \dots \text{Magnitude of resultant} \dots (1.1)$$

Now in $\triangle DOC$,

$$\tan \alpha = \frac{CD}{OD} = \frac{CD}{OA + AD}$$

$$\therefore \tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta} \dots \text{Direction of resultant} \dots (1.2)$$

Particular Cases

1. If $\theta = 90^\circ$, i.e. when the two forces act at right angle, then

$$R = \sqrt{P_1^2 + P_2^2} \text{ and } \tan \alpha = \frac{P_2}{P_1} \quad (\text{since } \cos 90^\circ = 0 \text{ and } \sin 90^\circ = 1)$$

2. If the two forces are equal, i.e. $P_1 = P_2$

$$\begin{aligned} R &= \sqrt{P_1^2 + P_1^2 + 2P_1^2 \cos \theta} = \sqrt{2P_1^2 (1 + \cos \theta)} \\ &= \sqrt{2P_1^2 \times 2\cos^2 \frac{\theta}{2}} \quad (\text{since } 1 + \cos \theta = 2\cos^2 \frac{\theta}{2}) \\ &= \sqrt{4P_1^2 \cos^2 \frac{\theta}{2}} \\ \therefore R &= 2P_1 \cos \frac{\theta}{2} \end{aligned}$$

Triangle Law of Forces (Graphical method)

It state, "If two forces acting simultaneously at a point be represented in magnitude and direction by two sides of triangle taken in order, their resultant may be represented in magnitude and direction by the third side of triangle taken in opposite order"

Triangle law is the equivalent statement of parallelogram law, which is shown in figure 1.5; this figure is one-half of the parallelograms. Here the tail of P_2 is placed at the tip of P_1 and R (resultant) is the vector that completes the triangle, drawn from the tail of P_1 to tip the tip of P_2 . The result is identical if the tail of P_1 is placed at the tip of P_2 and R is drawn from the tail of P_2 to the tip of P_1 .

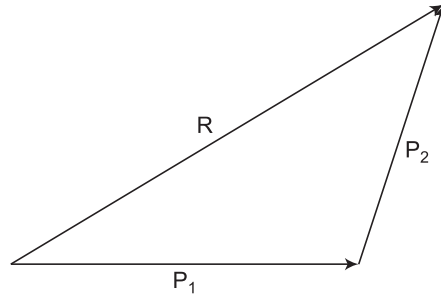


Fig. 1.5

CASE II. Resultant of more than two concurrent forces:

If three or more forces are acting at a point then their resultant is found analytically by a method known as rectangular components method. In this method all the forces acting at a point are resolved (resolution of force is discussed in detail in next section) into horizontal and vertical components and then algebraic sum of horizontal components (written as ΣH) and vertical components (written as ΣV) found separately.

Then the resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2} \dots \text{Magnitude of resultant} \quad \dots(1.3)$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H} \dots \text{Direction of resultant} \quad \dots(1.4)$$

Polygon Law of forces (Graphical method)

The resultant of three or more forces acting at a point is found graphically by polygon law of forces, it state,

“If a number of forces acting simultaneously at a point be represented in magnitude and direction by the sides of a polygon taken in order, their resultant may be represented in magnitude and direction by the closing side of the polygon taken in opposite order”

If four forces P_1 , P_2 , P_3 and P_4 are acting at a point as shown in figure 1.6, then the resultant of these forces is obtained by drawing a polygon of forces as explained below and shown in figure 1.7.

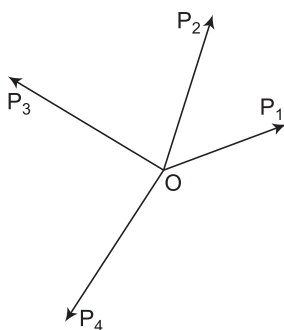


Fig. 1.6

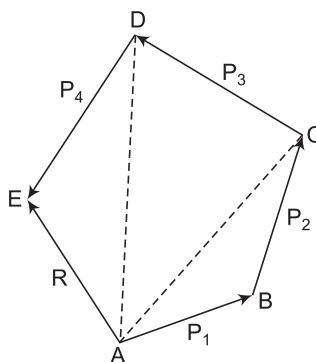


Fig. 1.7

With suitable scale from any convenient point A , draw AB parallel and equal to P_1 . From B draw BC parallel and equal to P_2 . From C draw CD parallel and equal to P_3 . From D draw DE parallel and equal to P_4 . Draw AE from the tail of P_1 to the tip of P_4 , then AE represents resultant of four forces P_1, P_2, P_3 and P_4 . Also AC represents resultant of P_1 and P_2 and AD represents resultant of AC and CD . $ABCDEA$ is the force polygon and it can be extended to any number of forces.

1.4 RESOLUTION OF A FORCE

Any force can be resolved into individual component forces in the same way as individual component forces can be composed together. It is often convenient to decompose a single force into two distinct forces; they are known as **components**. These forces, when acting together, have the same external effect on a body as the original force.

Resolution is the process of splitting up the given force into components, without changing its effect on the body

There are mainly two methods of resolving a force.

1. Perpendicular resolution
2. Non-perpendicular resolution

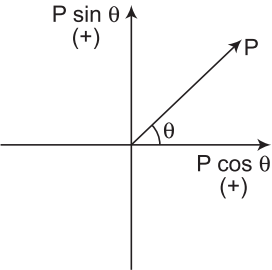
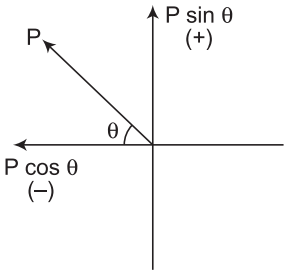
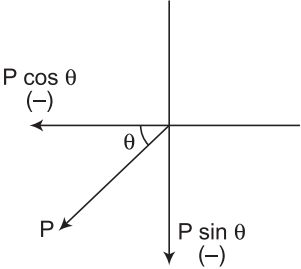
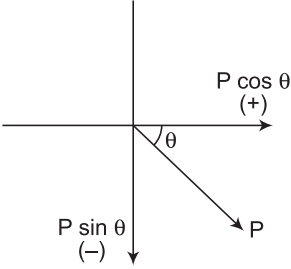
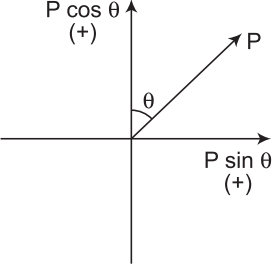
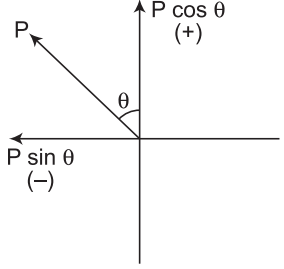
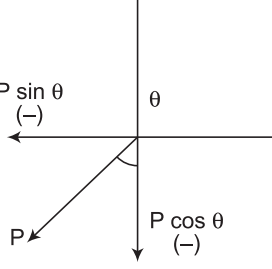
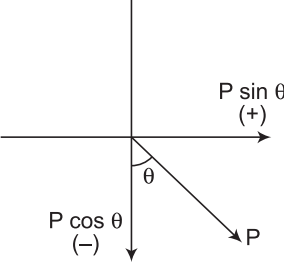
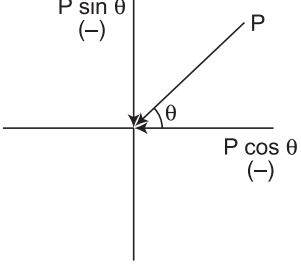
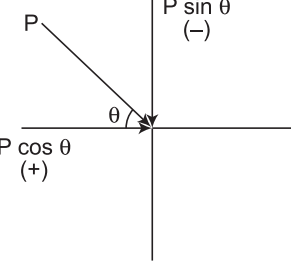
1. **Perpendicular Resolution:** Perpendicular resolution is the general method of splitting up a single force into two mutually perpendicular directions. Two perpendicular components are acting along x -axis and y -axis or any two perpendicular axes.

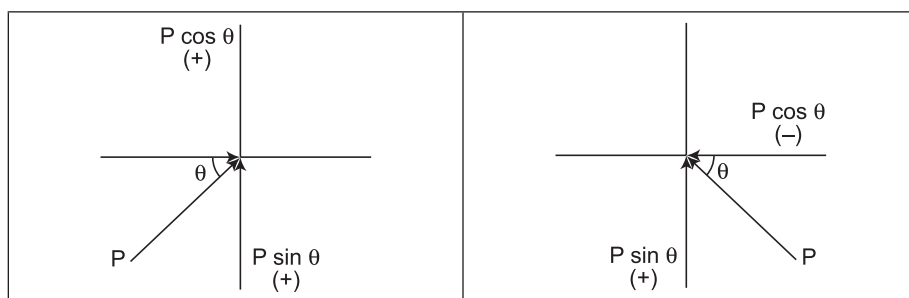
Table 1.2 illustrates the resolution of a force into perpendicular components.

Following sign conventions are used in drawing components,

- (a) Components acting horizontally towards right are positive and components acting horizontally towards left are negative.
- (b) Components acting vertically upward are positive and components acting vertically downwards are negative.

Table 1.2



When body is placed on an inclined plane, its weight 'mg' can be resolved into two components i.e. parallel to incline plane and perpendicular to incline plane. Here we can select x -axis along the plane and y -axis perpendicular to plane as shown in *figure 1.8*.

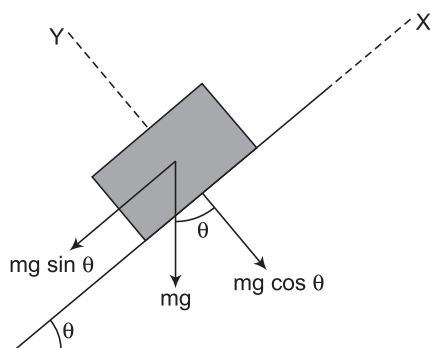


Fig. 1.8

- 2. Non-Perpendicular Resolution:** In non-perpendicular resolution a force is resolved into two directions which are not perpendicular to each other. If force P is to be resolved into two non-perpendicular directions let say along OA and OB as shown in *figure 1.9*, where OA and OB are not perpendicular to each other. Now construct a parallelogram by keeping two components (i.e. components along OA and OB) along two adjacent sides of parallelogram and original given force along the diagonal of parallelogram.

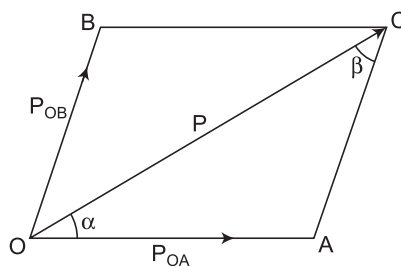


Fig. 1.9

Now apply sine rule in triangle OAC ,

$$\frac{P_{OB}}{\sin \alpha} = \frac{P_{OA}}{\sin \beta} = \frac{P}{\sin[180 - (\alpha + \beta)]}$$

$$\therefore P_{OA} = \frac{P \sin \beta}{\sin[180 - (\alpha + \beta)]} \dots \text{This is component of force } P \text{ along } OA$$

$$\therefore P_{OB} = \frac{P \sin \alpha}{\sin[180 - (\alpha + \beta)]} \dots \text{This is component of force } P \text{ along } OB$$

1.5 MOMENT

When a force causes an object to turn, this turning effect is called moments. So the moment of a force is a measure of its tendency to cause a body to rotate around a specific point or axis. Moment is the product of the force multiplied by the perpendicular distance from the line of action of the force to the pivot or point where the object will turn. The point about which moment is taken is known as moment center and the perpendicular distance is known as moment arm, as shown in *figure 1.10*. If moment arm d is zero, i.e. the line of action of the force P is passing through the moment center then moment due to this force will be zero.

The rotational effect produced by force is known as moment of the force

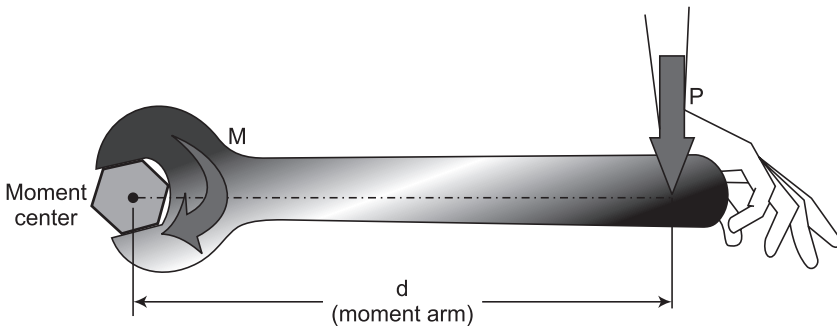


Fig. 1.10

$$\text{Moment} = \text{Force} \times \text{Distance}$$

$$M = F \times d \quad \dots(1.5)$$

Since moment is product of force and perpendicular distance, its S.I. unit is N-m or KN-m.

The direction of the rotation resulting from a moment is either clockwise or anticlockwise. When force causes an object to turn in clockwise direction, it is called clockwise moment and when force causes an object to turn in anticlockwise direction, it is called anticlockwise moment. Anticlockwise moments are regarded as positive, while clockwise moments are negative. (We shall assume this sign convention throughout this book)

Anticlockwise moment $\longrightarrow$ Positive 

Clockwise moment $\longrightarrow$ Negative 

1.5.1 Resultant Moment

When two or more than two forces are acting about a point their combined effect is represented by the “Resultant Moment”. To find out the resultant moment about a point, first find the sum of the anticlockwise moments and clockwise moments about the point, then take the lesser of these two moments from the greater and the difference is the magnitude of the resultant moment. The direction of the resultant moment will be that of the greater of the two component moments.

1.5.2 Varignon’s Theorem (Principle of Moment)

The Varignon theorem is a theorem given by French mathematician Pierre Varignon in 1687. It states, “*The moment of resultant of a force system about any point is equal to the algebraic sum of moments of all other forces about the same point*”.

Let R be the resultant of two concurrent forces P_1 and P_2 . Let R , P_1 and P_2 make angles θ , α and β with axis respectively as shown in figure 1.11. The moment of R about an arbitrary point A is, $R \times d$, where d is the perpendicular distance from line of action of R to point A . Similarly if we consider d_1 and d_2 are perpendicular distances from line of action of P_1 and P_2 to point A , then the sum of moments of P_1 and P_2 about point A are $P_1 \times d_1 + P_2 \times d_2$.

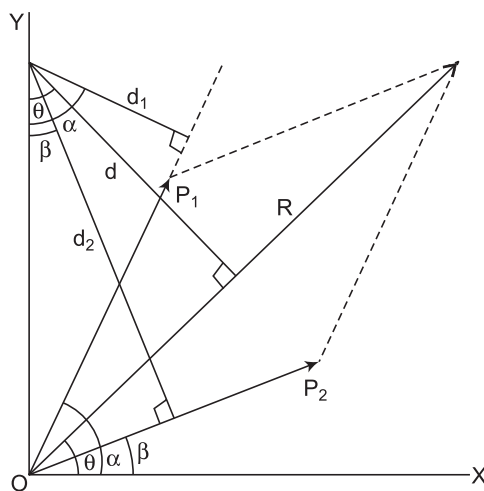


Fig. 1.11

Now, the sum of vertical component of P_1 and P_2 is equal to the vertical component of their resultant R . (As R is the resultant of P_1 and P_2)

i.e.,
$$R_y = P_{1y} + P_{2y}$$

But, from figure, $R_y = R \cos \theta$, $P_{1y} = P_1 \cos \alpha$ and $P_{2y} = P_2 \cos \beta$

$$\therefore R \cos \theta = P_1 \cos \alpha + P_2 \cos \beta$$

Now multiply on both sides by OA

$$R \times OA \cos \theta = P_1 \times OA \cos \alpha + P_2 \times OA \cos \beta$$

But, from figure, $OA \cos \theta = d$, $OA \cos \alpha = d_1$ and $OA \cos \beta = d_2$

$$\therefore R \times d = P_1 \times d_1 + P_2 \times d_2 \quad \dots(1.6)$$

Hence, it is proved that, the moment of resultant of a force system about any point is equal to the algebraic sum of moments of all other forces about the same point.

1.5.3 Resultant of Parallel Force System

Case I: Like Parallel Forces

When two parallel forces are acting in same direction they are known as like parallel forces. Their resultant can be find out using following steps, consider figure 1.12.

1. Find resultant $R = P_1 + P_2$
2. Take the algebraic sum moment of forces about point O ,
 Moment of P_1 about $O = P_1 \times AO$ (clockwise moment)
 Moment of P_2 about $O = P_2 \times BO$ (anticlockwise moment)
3. Position of resultant R can be obtained by applying Varignon's theorem about point O .

$$-P_1 \times AO + P_2 \times BO = R \times CO$$

In above equation, CO is the perpendicular distance between line of action of resultant R and reference point O .

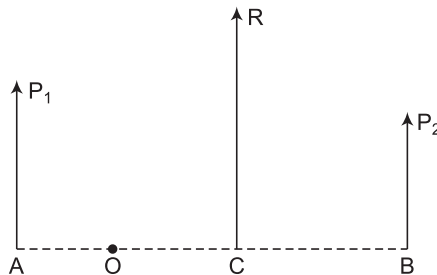


Fig. 1.12

Case II: Unlike Parallel Forces

When two parallel forces are acting in opposite direction they are known as unlike parallel forces. Their resultant can be find out using following steps, consider figure 1.13.

1. Find resultant $R = P_1 - P_2$
2. Take the algebraic sum moment of forces about point O ,
 Moment of P_1 about $O = P_1 \times AO$ (clockwise moment)
 Moment of P_2 about $O = P_2 \times BO$ (clockwise moment)

3. Position of resultant R can be obtained by applying Varignon's theorem about point O .

$$-P_1 \times AO - P_2 \times BO = -R \times CO$$

In above equation, CO is the perpendicular distance between line of action of resultant R and reference point O .

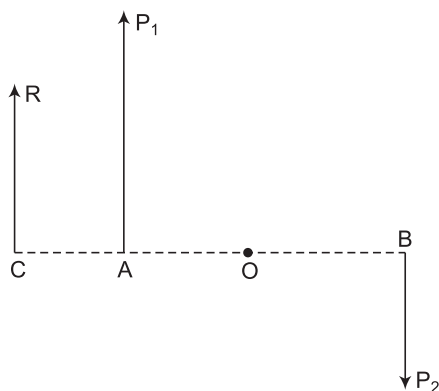


Fig. 1.13

1.6 COUPLE

A couple consists of a pair of two forces which has the following properties:

- Equal magnitude and opposite in direction
- Act along parallel lines of action
- Separated by a perpendicular distance d , known as arm of couple. A couple is shown in *figure 1.14*.

Two equal, unlike parallel, non-collinear forces form a couple

Since the two forces cancel out each other giving zero resultant, a couple produces purely rotational effect in the body without translation motion. A couple cannot be balanced by a single force but it can be balanced only by another couple of opposite nature.

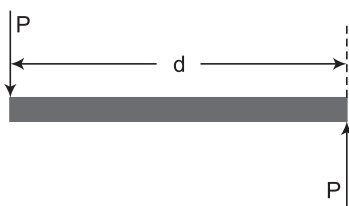


Fig. 1.14

1.6.1 Moment of a Couple

Consider a couple as shown in *figure 1.15*.

The moment of couple about point *A* is,

$$M_A = P \times d_2 - P \times d_1$$

$$M_A = P (d_2 - d_1)$$

But from *figure 1.15*, $d_2 - d_1 = d$

$$M_A = P \times d$$

The moment of a couple is the product of magnitude of one of the forces and arm of couple.

Now consider point *B* at distance $\frac{d}{2}$ from force *P*, the moment of couple about point *B* is,

$$M_B = P \times \frac{d}{2} + P \times \frac{d}{2}$$

$$M_B = P \left(\frac{d}{2} + \frac{d}{2} \right)$$

$$M_B = P \times d \quad \dots(1.7)$$

From above we see that, the moment of couple about any pivot point is $P \times d$.

It is important to note here that a couple does not have moment center, like moment of force. A couple has the same moment about all points on a body.

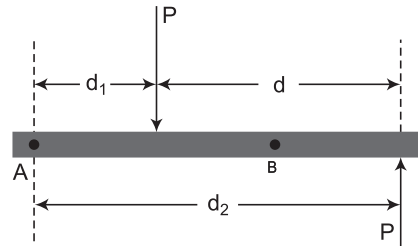


Fig. 1.15

1.6.2 Force–Couple System

The concept of couple is useful in application of parallel transfer of a force.

Refer to *figure 1.16*, consider a force *P* is acting at point *A* and it is required to transfer from point *A* to point *B* without changing its magnitude and direction.

Now apply two forces of equal magnitude (their magnitude are same as that of force *P* acting at point *A*) and opposite in direction at point *B*, as shown in *figure 1.17*.



Fig. 1.16

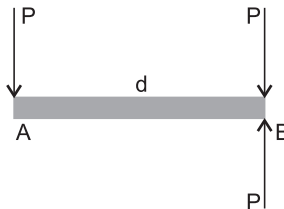


Fig. 1.17

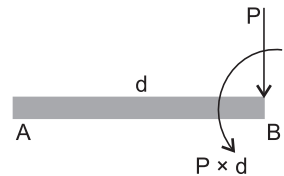


Fig. 1.18

Observing *figure 1.17*, we see that two forces are acting in opposite direction at point *A* and *B* form a couple. Moment of this couple is $P \times d$, anticlockwise. Thus, to shift a force to point *B*, a couple is required to be added to the system.

This system which consisting a force and a couple at point B , as shown in figure 1.18 is known as force-couple system. So we can transfer a force parallel to itself at any point with force-couple system.

It is also possible to replace a force-couple system into a single force by following exactly the reverse procedure as discuss above.

1.7 EQUIVALENT SYSTEMS OF FORCES

Two force systems that produce the same external effects on a rigid body are said to be equivalent, i.e. the sums of the forces and sums of the moments about a point are equal. Equivalent system consists of a single resultant force R through the given point and single resultant moment.

Two systems are equivalent only when,

1. The sum of the forces in system 1 is equal to the sum of the forces in system 2
2. The sum of the moments about any point O in system 1 is equal to the sum of the moments about same point O in system 2.

Solved Examples Based on Parallelogram Law of Forces

Example 1. Two forces 20 N and 25 N are acting at a point with an angle of 60° between them. Find the magnitude and direction of resultant.

Solution: Here $P_1 = 20$ N, $P_2 = 25$ N and $\theta = 60^\circ$

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{20^2 + 25^2 + 2 \times 20 \times 25 \cos 60}$$

$$R = 39.05 \text{ N} \quad \dots \text{Magnitude of resultant} \quad (\text{Ans.})$$

Now we know that

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{25 \sin 60}{20 + 25 \cos 60}$$

$$\therefore \alpha = 33.66^\circ \quad \dots \text{Direction of resultant} \quad (\text{Ans.})$$

Example 2. Force P_1 of magnitude 10 N is acting along horizontal direction and force P_2 acting along vertical direction. The resultant of P_1 and P_2 has magnitude of 20 N. Determine the magnitude of force P_2 and direction of resultant.

Solution: Here $P_1 = 10$ N, $R = 20$ N and $\theta = 90^\circ$

As we know that when $\theta = 90^\circ$ in parallelogram then,

$$R = \sqrt{P_1^2 + P_2^2}$$

$$\begin{aligned}\therefore 20 &= \sqrt{10^2 + P_2^2} \\ \therefore 20^2 &= 10^2 + P_2^2 \\ \therefore P_2 &= 17.32 \text{ N (Ans.)}\end{aligned}$$

Now we also know that when $\theta = 90^\circ$ in parallelogram then

$$\begin{aligned}\tan \alpha &= \frac{P_2}{P_1} \\ \therefore \tan \alpha &= \frac{17.32}{10} \\ \therefore \alpha &= 59.5^\circ \text{ (Ans.)}\end{aligned}$$

Example 3. Two forces acting at point and angle between them is 120° . The larger force is of magnitude 80 N and the resultant of these two forces is perpendicular to the smaller force. Find magnitude of smaller force and resultant.

Solution: Here, $P_1 = 80 \text{ N}$, $\theta = 120^\circ$, and angle between smaller force i.e. P_2 and resultant is 90° .

As we know from parallelogram law of forces,

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

In above equation $\alpha = 30^\circ$.

$$\begin{aligned}\therefore \tan 30^\circ &= \frac{P_2 \sin 120}{80 + P_2 \cos 120} \\ \therefore 0.5773 &= \frac{P_2 0.8660}{80 + P_2 (-0.50)} = \frac{P_2 0.8660}{80 - 0.5 P_2} \\ \therefore 80 - 0.5 P_2 &= 1.5 P_2 \\ \therefore 2 P_2 &= 80 \\ \therefore P_2 &= 40 \text{ N (Ans.)}\end{aligned}$$

Now from parallelogram law of forces,

$$\begin{aligned}R &= \sqrt{P_1^2 + P_2^2 + 2 P_1 P_2 \cos \theta} \\ \therefore R &= \sqrt{80^2 + 40^2 + 2 \times 80 \times 40 \cos 120} \\ \therefore R &= 69.28 \text{ N (Ans.)}\end{aligned}$$

Example 4. The maximum resultant of two forces P_1 and P_2 is 1000 N and minimum magnitude is 400 N. Find values of P_1 and P_2 .

Solution: The resultant of two forces will be maximum only when the two forces will act in same direction as shown in *Figure Ex. 4 (a)* and resultant is minimum only when the two forces will act in opposite direction as shown in *Figure Ex. 4(b)*.



Fig. Ex. 4(a)



Fig. Ex. 4(b)

Now from *figure Ex. 4(a)*, angle between two forces $\theta = 0$

From parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

Substituting $\theta = 0$

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2}$$

$$\therefore R = \sqrt{(P_1 + P_2)^2}$$

$$\therefore R = P_1 + P_2$$

$$\therefore P_1 + P_2 = 1000 \quad \dots(\text{Maximum resultant}) \quad \dots(\text{Eq. 1})$$

Now from *figure Ex. 4(b)*, angle between two forces $\theta = 180^\circ$

From parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

Substituting $\theta = 180^\circ$

$$R = \sqrt{P_1^2 + P_2^2 - 2P_1P_2}$$

$$\therefore R = \sqrt{(P_1 - P_2)^2}$$

$$\therefore R = P_1 - P_2$$

$$\therefore P_1 - P_2 = 400 \quad \dots(\text{Minimum resultant}) \quad \dots(\text{Eq. 2})$$

Solving Eq. 1 and Eq. 2, we get

$$P_1 = 700 \text{ N and } P_2 = 300 \text{ N} \quad (\text{Ans.})$$

Example 5. Two forces of 375 N and 250 N are acting on a hook and passing through point *O* as shown in *figure Ex. 5*. Determine magnitude and direction of resultant of these two forces.

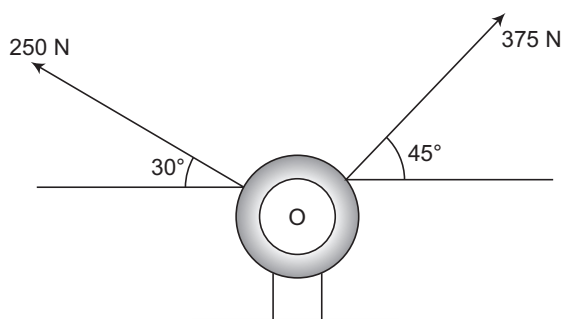


Fig. Ex. 5

Solution: Here, $P_1 = 375$ N, $P_2 = 250$ N and $\theta = 105^\circ$.

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{375^2 + 250^2 + 2 \times 375 \times 250 \cos 105^\circ}$$

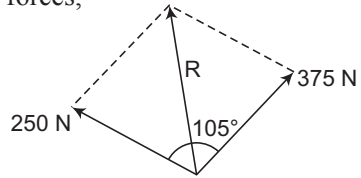
$$\therefore R = 393.18 \text{ N} \quad \dots \text{Magnitude of resultant (Ans.)}$$

Now, as we know from parallelogram law of forces,

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{250 \sin 105^\circ}{375 + 250 \cos 105^\circ}$$

$$\therefore \alpha = 37.9^\circ \quad \dots \text{Direction of resultant (Ans.)}$$



Example 6. Find the magnitude of the two forces, such that if they act at right angles, their resultant is 5 N but when they act at 60° , their resultant is $\sqrt{37}$ N.

Solution: First consider two forces P_1 and P_2 acting at right angle,

$\therefore$ angle between two forces $\theta = 90^\circ$.

As we know that when $\theta = 90^\circ$ in parallelogram then

$$R = \sqrt{P_1^2 + P_2^2}$$

$$\therefore 5 = \sqrt{P_1^2 + P_2^2}$$

$$\therefore 25 = P_1^2 + P_2^2 \quad \dots (\text{Eq. 1})$$

Similarly, when the angle between the two forces is 60° , then resultant force from parallelogram law is,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore \sqrt{37} = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos 60^\circ}$$

$$\therefore 37 = P_1^2 + P_2^2 + 2P_1P_2 \times 0.5$$

Substituting $P_1^2 + P_2^2 = 25$ from Eq. 1

$$\therefore 37 = 25 + 2P_1P_2 \times 0.5$$

$$\therefore P_1P_2 = 37 - 25 \quad \therefore P_1P_2 = 12 \quad \dots (\text{Eq. 2})$$

Now we know that,

$$(P_1 + P_2)^2 = P_1^2 + P_2^2 + 2P_1P_2$$

Substituting values from Eq. 1 and Eq. 2

$$(P_1 + P_2)^2 = 25 + 2 \times 12 = 49$$

$$\therefore P_1 + P_2 = \sqrt{49} = 7 \quad \dots (\text{Eq. 3})$$

Similarly, we know that,

$$(P_1 - P_2)^2 = P_1^2 + P_2^2 - 2P_1P_2$$

Substituting values from Eq. 1 and Eq. 2

$$(P_1 - P_2)^2 = 25 - 2 \times 12 = 1$$

$$\therefore P_1 - P_2 = \sqrt{1} = 1$$

... (Eq. 4)

Solving Eq. 3 and Eq. 4 we get,

$$P_1 = 4 \text{ N and } P_2 = 3 \text{ N (Ans.)}$$

Example 7. A disabled automobile is pulled using ropes subjected to the two forces as shown in *figure Ex. 7*. Determine the magnitude and direction of their resultant.

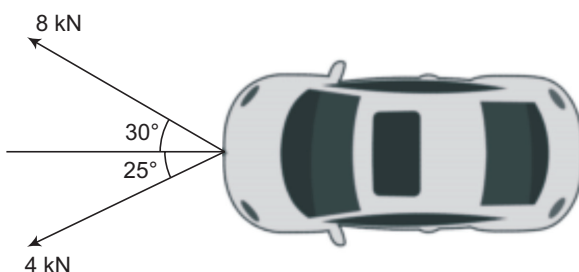
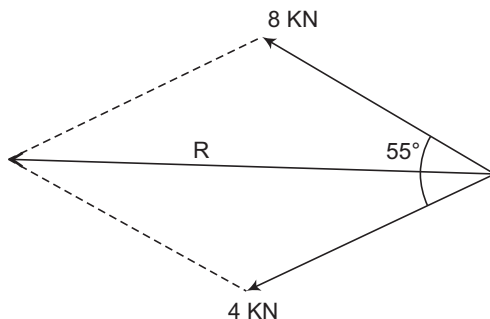


Fig. Ex. 7

Solution: Here, $P_1 = 4 \text{ kN}$, $P_2 = 8 \text{ kN}$ and $\theta = 55^\circ$



As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{4^2 + 8^2 + 2 \times 4 \times 8 \cos 55^\circ}$$

$$\therefore R = 10.80 \text{ kN} \quad \dots \text{Magnitude of resultant (Ans.)}$$

Similarly we know from parallelogram law of forces,

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\tan \alpha = \frac{8 \sin 55^\circ}{8 + 4 \cos 55^\circ}$$

$$\alpha = 32.49^\circ \quad \dots \text{Direction of resultant (Ans.)}$$

Example 8. Two forces of 240 N and 200 N are acting at point O as shown in figure Ex. 8. Determine magnitude of resultant and direction of resultant from positive x -axis.

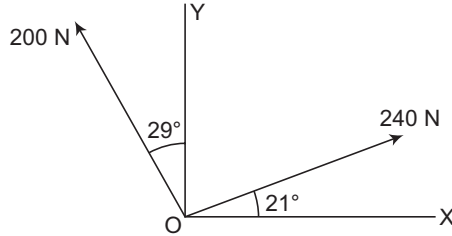


Fig. Ex. 8

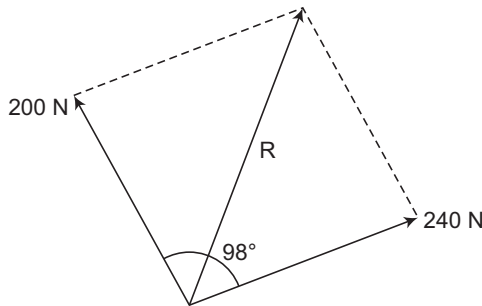
Solution: Here, $P_1 = 240 \text{ N}$, $P_2 = 200 \text{ N}$ and $\theta = 90^\circ - 21^\circ + 29^\circ = 98^\circ$.

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{240^2 + 200^2 + 2 \times 240 \times 200 \cos 98^\circ}$$

$$\therefore R = 290.24 \text{ N} \quad \dots \text{Magnitude of resultant} \quad \textbf{(Ans.)}$$



Now, as we know from parallelogram law of forces,

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{200 \sin 98^\circ}{240 + 200 \cos 98^\circ}$$

$$\therefore \alpha = 43.02^\circ$$

Now direction of resultant from positive x axis is

$$21^\circ + 43.02^\circ = 64.02^\circ \quad \dots \text{Direction of resultant} \quad \textbf{(Ans.)}$$

Example 9. Two forces of 600 N and 700 N are acting at point O as shown in figure Ex. 9. Determine magnitude of resultant and direction of resultant from x -axis.

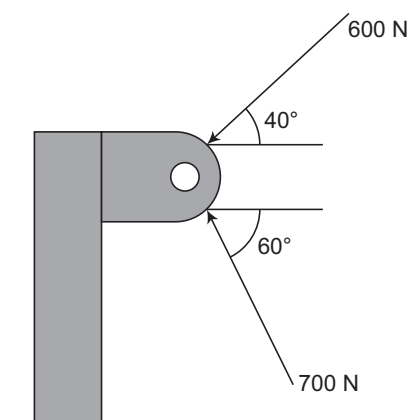


Fig. Ex. 9

Solution: Here, $P_1 = 700$ N, $P_2 = 600$ N and $\theta = 100^\circ$.

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{700^2 + 600^2 + 2 \times 700 \times 600 \cos 100^\circ}$$

$$\therefore R = 839.12 \text{ N}$$

...Magnitude of resultant **(Ans.)**

Now, as we know from parallelogram law of forces,

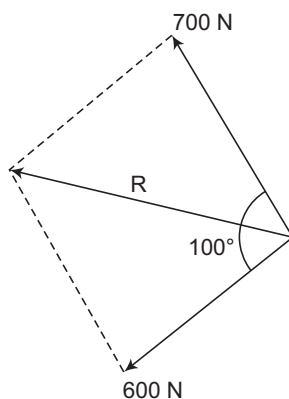
$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{600 \sin 100^\circ}{700 + 600 \cos 100^\circ}$$

$$\therefore \alpha = 44.76^\circ$$

Now direction of resultant from x axis is

$$60^\circ - 44.76^\circ = 15.24^\circ \quad \textbf{(Ans.)}$$



Example 10. Two forces of 600 N and 500 N are acting at point O as shown in figure Ex. 10. Determine magnitude of resultant and direction of resultant from x axis.

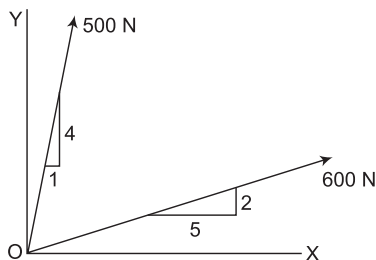


Fig. Ex. 10

Solution: Here, $P_1 = 600 \text{ N}$, $P_2 = 500 \text{ N}$ and $\theta = \tan^{-1} \frac{4}{1} - \tan^{-1} \frac{2}{5} = 54.16^\circ$

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{600^2 + 500^2 + 2 \times 600 \times 500 \cos 54.16^\circ}$$

$$\therefore R = 980.47 \text{ N} \quad \dots \text{Magnitude of resultant (Ans.)}$$

Now, as we know from parallelogram law of forces,

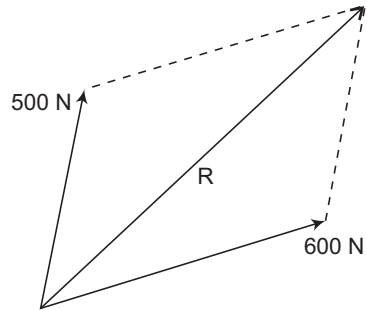
$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{500 \sin 54.16^\circ}{600 + 500 \cos 54.16^\circ}$$

$$\therefore \alpha = 24.41^\circ$$

Now direction of resultant from x -axis is

$$24.41^\circ + \tan^{-1} \frac{2}{5} = 46.22^\circ \quad (\text{Ans.})$$



Example 11. Two forces of 82.5 kN and 74 kN are acting on a bracket as shown in figure Ex. 11. Determine magnitude of resultant and direction of resultant from x -axis.

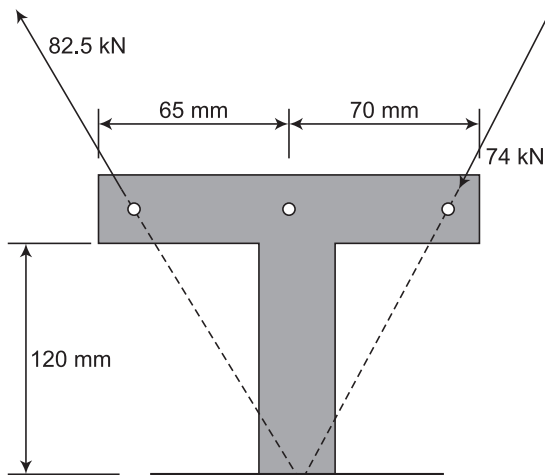


Fig. Ex. 11

Solution: Here, $P_1 = 82.5 \text{ kN}$, $P_2 = 74 \text{ kN}$

$$\text{and } \theta = 180^\circ - \tan^{-1} \frac{65}{120} - \tan^{-1} \frac{70}{120} = 121.30^\circ$$

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{82.5^2 + 74^2 + 2 \times 82.5 \times 74 \cos 121.30^\circ}$$

$$\therefore R = 77.06 \text{ kN} \quad \dots \text{Magnitude of resultant (Ans.)}$$

Now, as we know from parallelogram law of forces,

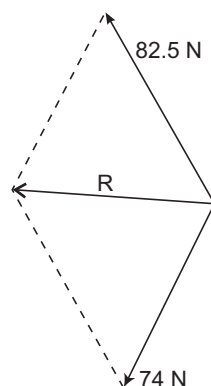
$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{74 \sin 121.30^\circ}{82.5 + 74 \cos 121.30^\circ}$$

$$\therefore \alpha = 55.13^\circ$$

Now direction of resultant from x -axis is

$$55.13^\circ + \tan^{-1} \frac{65}{120} = 83.6^\circ \quad (\text{Ans.})$$



Solved Examples Based on Resolution of Forces

Example 1. Resolve 100 N force into components along (a) x and y directions and (b) u and v directions. Refer figure Ex. 1

Solution: (a) Components along x and y :

Component along $x = 100 \cos 40^\circ = 76.60 \text{ N}$ (Ans.)

Component along $y = 100 \sin 40^\circ = 64.28 \text{ N}$ (Ans.)

(b) Component along u and v :

As angle between u and v is 100° , so this is non-perpendicular resolution.

Construct parallelogram by considering 100 N force as a resultant and its component along u and v as shown in figure Ex. 1(a)

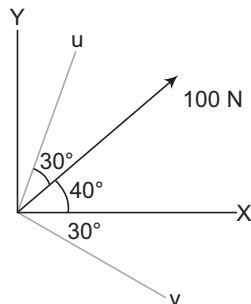


Fig. Ex. 1

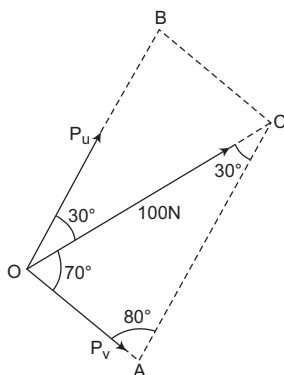


Fig. Ex. 1(a)

Now in triangle OAC , apply sine rule

$$\frac{100}{\sin 80^\circ} = \frac{P_u}{\sin 70^\circ} = \frac{P_v}{\sin 30^\circ}$$

$$P_u = \frac{100 \sin 70^\circ}{\sin 80^\circ} = 95.42 \text{ N} \quad (\text{Ans})$$

$$P_v = \frac{100 \sin 30^\circ}{\sin 80^\circ} = 50.77 \text{ N} \quad (\text{Ans})$$

Example 2. If the force P as shown in figure Ex. 2 is resolved into components parallel to the bars AB and BC , the magnitude of the component parallel to bar BC is 4 kN. What is the magnitudes of P and its component along AB ?

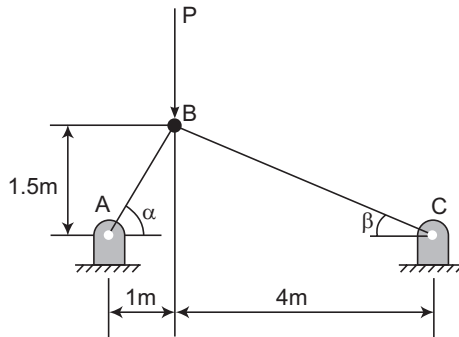


Fig. Ex. 2

Solution: Here taking a force P as a resultant, construct a parallelogram taking two side of parallelogram as two component of force P along AB and BC as shown in figure Ex. 2(a).

Now, from figure Ex. 2

$$\tan \alpha = \frac{1.5}{1.0} \quad \therefore \alpha = 56.31^\circ$$

$$\tan \beta = \frac{1.5}{4.0} \quad \therefore \beta = 20.56^\circ$$

From figure Ex. 2(a)

$$\theta = 90 - \alpha \quad \therefore \theta = 90 - 56.31^\circ = 33.69^\circ$$

$$\Phi = 90 - \beta \quad \therefore \Phi = 90 - 20.56^\circ = 69.54^\circ$$

Now apply sine rule to figure Ex. 2(a)

$$\frac{P}{\sin (180 - \theta - \Phi)} = \frac{P_{AB}}{\sin \Phi} = \frac{P_{BC}}{\sin \theta}$$

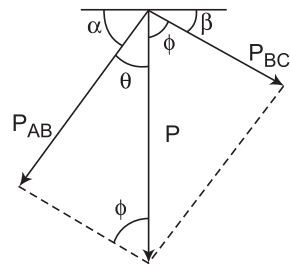


Fig. Ex. 2(a)

$$\frac{P}{\sin(180 - 33.69 - 69.54)} = \frac{4}{\sin 33.69}$$

$$\therefore P = \frac{4 \sin 76.87}{\sin 33.69} \quad \therefore P = 7 \text{ kN (Ans)}$$

$$\frac{P_{AB}}{\sin 69.54} = \frac{4}{\sin 33.69}$$

$$\therefore P_{AB} = \frac{4 \sin 69.54}{\sin 33.69} \quad \therefore P_{AB} = 6.75 \text{ kN (Ans)}$$

Example 3. A force P is inclined at 60° to the horizontal. If the horizontal component of force is 50 N find out vertical component of force.

Solution: Here the force P is inclined to the horizontal at an angle $\theta = 60^\circ$.

Horizontal component of force $P = P \cos \theta$

$$P \cos 60^\circ = 50 \quad \therefore P = \frac{50}{\cos 60^\circ} = 100 \text{ N}$$

Now vertical component of $P = P \sin \theta = 100 \sin 60^\circ = 86.60 \text{ N (Ans.)}$

Example 4. A box is to be moved in the direction of OO' as shown in figure Ex. 4. It is identified that a 20 kN force in this direction is required but obstruction prevent direct application of such a force. Accordingly, the forces P_1 and P_2 are applied as shown. Determine the magnitude of P_1 and P_2 .

Solution: Here it is desired that 20 kN force in the direction of OO' is the resultant of P_1 and P_2 .

Now the parallelogram may be constructed taking a 20 kN force as a resultant and by drawing lines parallel to the specified direction of P_1 and P_2 as shown in figure Ex. 4(a).

Now in triangle OAC , apply sine rule

Figure Ex. 4(a)

$$\frac{20}{\sin 45^\circ} = \frac{P_1}{\sin 60^\circ} = \frac{P_2}{\sin 75^\circ}$$

$$\therefore P_1 = \frac{20 \sin 60^\circ}{\sin 45^\circ} = 24.5 \text{ kN (Ans)}$$

$$\therefore P_2 = \frac{20 \sin 75^\circ}{\sin 45^\circ} = 27.32 \text{ kN (Ans)}$$

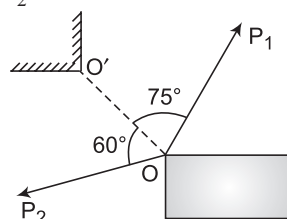


Fig. Ex. 4

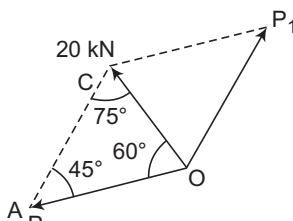


Fig. Ex. 4(a)

Example 5. Crank ABC is subjected to force P as shown in figure Ex. 5. Knowing that P must have a 200 N component perpendicular to arm AB of the crank, determine the magnitude of force P and its component along AB .

$$\therefore P_{AB} = \frac{50 \sin 60^\circ}{\sin 75^\circ} = 44.82 \text{ N (Ans)}$$

$$\therefore P_{AC} = \frac{50 \sin 45^\circ}{\sin 75^\circ} = 36.6 \text{ N (Ans)}$$

Example 7. Resolve 800 N force into components along u and v directions as shown in figure Ex. 7.

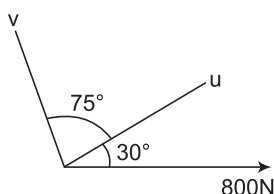


Fig. Ex. 7

Solution: Construct parallelogram by considering 800 N force as a resultant and its component along u and v as shown in figure Ex. 7(a).

Now in triangle OAC , apply sine rule

$$\frac{800}{\sin 75^\circ} = \frac{P_u}{\sin 75^\circ} = \frac{P_v}{\sin 30^\circ}$$

$$\therefore P_u = \frac{800 \sin 75^\circ}{\sin 75^\circ} = 800 \text{ N (Ans)}$$

$$\therefore P_v = \frac{800 \sin 30^\circ}{\sin 75^\circ} = 414.11 \text{ N (Ans)}$$

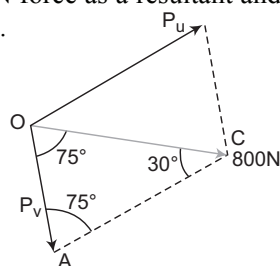


Fig. Ex. 7(a)

Example 8. Two forces P_1 and P_2 are applied to a hook as shown in figure Ex. 8. The resultant of the two forces has a magnitude of 300 N and makes an angle of 30° with positive x -axis (anticlockwise direction). Determine the magnitudes of P_1 and P_2 .

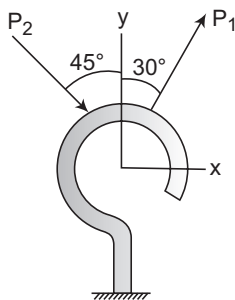


Fig. Ex. 8

Solution: Here taking a 300 N force as a resultant, construct a parallelogram taking P_1 and P_2 as two adjacent sides of parallelogram as shown in *figure Ex. 8(a)*.

Now in triangle OAC , apply sine rule

$$\frac{300}{\sin 75^\circ} = \frac{P_1}{\sin 75^\circ} = \frac{P_2}{\sin 30^\circ}$$

$$\therefore P_1 = \frac{300 \sin 75^\circ}{\sin 75^\circ} = 300 \text{ N} \quad (\text{Ans})$$

$$\therefore P_2 = \frac{300 \sin 30^\circ}{\sin 75^\circ} = 155.3 \text{ N} \quad (\text{Ans})$$

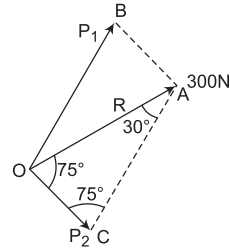


Fig. Ex. 8(a)

Example 9. A bracket is subjected to two forces P_1 and P_2 as shown in *figure Ex. 9*. If the resultant of the two forces has a magnitude of 525 N and direction as shown in the figure, determine magnitudes of P_1 and P_2 .

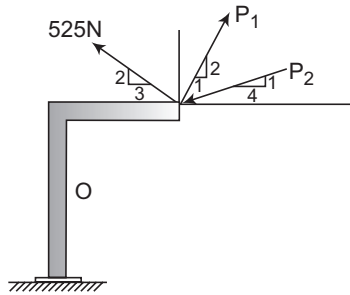


Fig. Ex. 9

Solution: Here first consider angle of P_2 , P_1 and resultant force are respectively θ_1 , θ_2 , and θ_3 from x-axis.

$$\text{Now } \theta_1 = \tan^{-1} \frac{1}{4} = 14.03^\circ \quad \theta_2 = \tan^{-1} \frac{2}{1} = 63.43^\circ \quad \theta_3 = \tan^{-1} \frac{2}{3} = 33.70^\circ$$

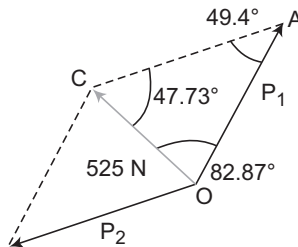


Fig. Ex. 9(a)

Now taking a 525 N force as a resultant, construct a parallelogram taking P_1 and P_2 as two adjacent sides of parallelogram as shown in *figure Ex. 9 (a)*.

Now in triangle OAC, apply sine rule

$$\frac{525}{\sin 49.4^\circ} = \frac{P_1}{\sin 47.73^\circ} = \frac{P_2}{\sin 82.87^\circ}$$

$$\therefore P_2 = \frac{525 \sin 82.87^\circ}{\sin 49.4^\circ} = 686.11 \text{ N}$$

$$\therefore P_1 = \frac{525 \sin 47.73^\circ}{\sin 49.4^\circ} = 511.66 \text{ N}$$

Example 10. A frame is subjected to a horizontal force of 200 N as shown in *figure Ex. 10*. If the component of force along AC is 250 N, directed from A towards C, determine the component of force along member AB and angle θ ($0^\circ \leq \theta \leq 90^\circ$) of member AB.

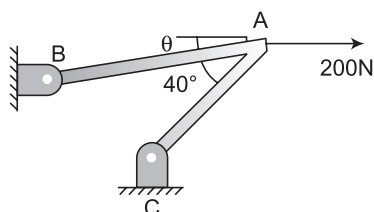


Fig. Ex. 10

Solution: Here taking a force 200 N as a resultant, construct a parallelogram taking two adjacent sides of parallelogram as two component of force along AC and AB as shown in *Figure Ex. 10(a)*

Now in triangle OAC, apply sine rule

$$\frac{200}{\sin 40^\circ} = \frac{250}{\sin \theta} = \frac{P_{AB}}{\sin (140 - \theta)^\circ}$$

$$\therefore \sin \theta = \frac{250 \sin 40^\circ}{200} = 0.8034$$

$$\therefore \theta = 53.46^\circ \quad (\text{Ans.})$$

$$\therefore P_{AB} = \frac{250 \sin (140 - 53.46)^\circ}{\sin 53.46^\circ} = 310.6 \text{ N} \quad (\text{Ans.})$$

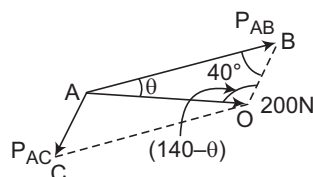


Fig. Ex. 10(a)

Example 11. A wooden log is to be hoisted using two chains as shown in *figure Ex. 11*. If the resultant force is to be 1200 N in vertical upward direction determine the magnitudes of P_1 and P_2 acting on each chain and angle θ of P_2 so that magnitude of P_2 is to be minimum.

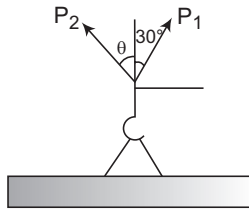


Fig. Ex. 11

Solution: Here taking a force 1200 N as a resultant, construct a parallelogram taking two adjacent sides of parallelogram as two component of force along AC and AB as shown in figure Ex. 11 (a).

Now in triangle OAC, apply sine rule

$$\frac{1200}{\sin(150 - \theta)^\circ} = \frac{P_1}{\sin \theta} = \frac{P_2}{\sin 30^\circ}$$

$$\therefore P_2 = \frac{1200 \sin 30^\circ}{\sin(150 - \theta)^\circ}$$

Now, for P_2 to be minimum the denominator $\sin(150 - \theta)$ must be maximum.

$$\therefore (150 - \theta) = 90^\circ \quad \therefore \theta = 60^\circ$$

$$\therefore P_2 = \frac{1200 \sin 30^\circ}{\sin(150 - 60)^\circ} = 600 \text{ N} \quad (\text{Ans.})$$

$$\text{Now} \quad P_1 = \frac{P_2 \sin 60^\circ}{\sin 30^\circ} = \frac{600 \sin 60^\circ}{\sin 30^\circ} = 1039.23 \text{ N} \quad (\text{Ans.})$$

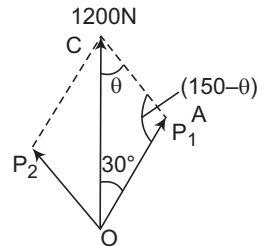


Fig. Ex. 11(a)

Example 12. The line of action of the 1000 N force is passing through the point A (-6, -2) and B (7,5) as shown in Figure Ex. 12. Determine the x and y components of 1000 N force.

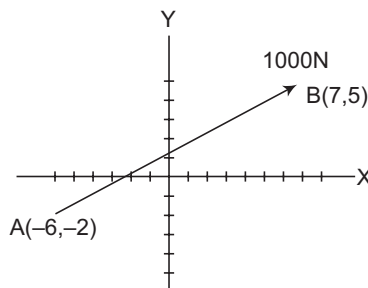


Fig. Ex. 12

Solution: Here angle made by 1000 N force with x axis is $\theta = \tan^{-1} \frac{7}{13} \therefore \theta = 28.3^\circ$

Now x component of 1000 N force = $1000 \cos 28.3^\circ = 880.48 \text{ N} \quad (\text{Ans.})$

y component of 1000 N force = $1000 \sin 28.3^\circ = 474 \text{ N} \quad (\text{Ans.})$

Example 13. Resolve 500 N force acting on bar into components along OA and perpendicular to OA . Refer figure Ex. 13.

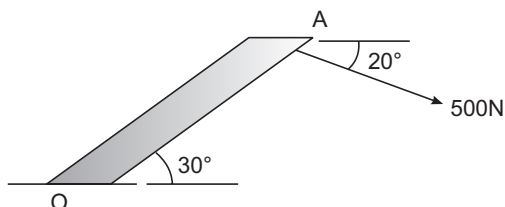


Fig. Ex. 13

Solution: The component of 500 N force along OA and perpendicular to OA are,
Component along $OA = 500 \cos 50^\circ = 321.4 \text{ N}$ (Ans.)

Component perpendicular to $OA = 500 \sin 50^\circ = 383 \text{ N}$ (Ans.)

Example 14. A frame is subjected to two forces as shown in figure Ex. 14. Replace them by equivalent forces along x and a direction.

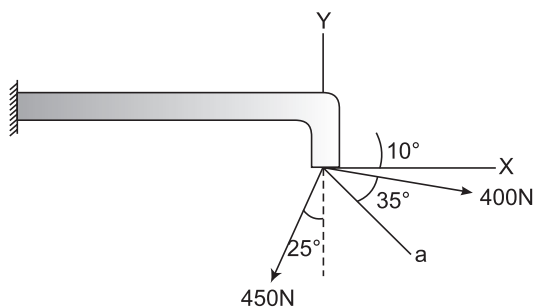


Fig. Ex. 14

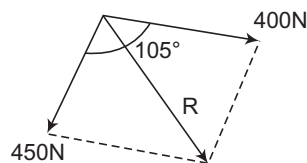


Fig. Ex. 14(a)

Solution: Here first find out resultant of given two forces using parallelogram law. Refer figure Ex. 14 (a)

Now take $P_1 = 450 \text{ N}$, $P_2 = 400 \text{ N}$ and angle θ between P_1 and $P_2 = 105^\circ$

As we know from parallelogram law of forces,

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta}$$

$$\therefore R = \sqrt{450^2 + 400^2 + 2 \times 450 \times 400 \cos 105^\circ}$$

$$\therefore R = 519 \text{ N}$$

Now, as we know from parallelogram law of forces,

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta}$$

$$\therefore \tan \alpha = \frac{400 \sin 105^\circ}{450 + 400 \cos 105^\circ} \quad \therefore \alpha = 48.11^\circ$$

Now taking a 519 N force as a resultant, construct a parallelogram taking two adjacent sides of parallelogram as two component of force along x and a direction as shown in figure Ex. 14(b).

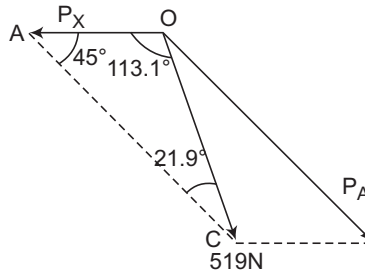


Fig. Ex.14 (b)

Now in triangle OAC , apply sine rule

$$\frac{519}{\sin 45^\circ} = \frac{P_a}{\sin 113.1^\circ} = \frac{P_x}{\sin 21.9^\circ}$$

$$\therefore P_a = \frac{519 \sin 113.1^\circ}{\sin 45^\circ} = 675.13 \text{ N} \quad (\text{Ans.})$$

$$\therefore P_x = \frac{519 \sin 21.9^\circ}{\sin 45^\circ} = 273.76 \text{ N} \quad (\text{Ans.})$$

Solved Examples Based on Co-planer Concurrent Forces

Example 1. Determine resultant of the force system shown in the figure Ex. 1

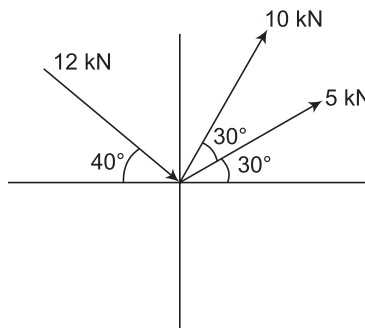


Fig. Ex. 1

Solution: Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components.

$$\begin{aligned} \rightarrow \Sigma H &= 5 \cos 30^\circ + 10 \cos 60^\circ + 12 \cos 40^\circ \\ &= 18.52 \text{ kN} \end{aligned}$$

$$\begin{aligned}
 +\uparrow \Sigma V &= 5 \sin 30^\circ + 10 \sin 60^\circ - 12 \sin 40^\circ \\
 &= 3.45 \text{ kN}
 \end{aligned}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(18.52)^2 + (3.45)^2}$$

$$\therefore R = 18.84 \text{ kN} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{3.45}{18.52} \qquad \theta = 10.55^\circ \quad (\text{Ans.})$$

Example 2. A hook is subjected to three forces as shown in the figure Ex. 2. Determine the magnitude of the resultant force and its direction from x -axis.

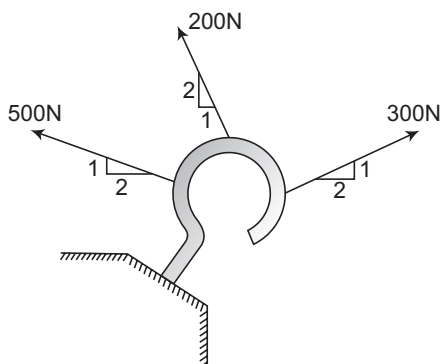


Fig. Ex. 2

Solution: Consider θ_1 , θ_2 and θ_3 are the angles of forces 300 N, 200 N and 500 N respectively.

$$\text{Now,} \quad \theta_1 = \tan^{-1} \frac{1}{2} = 26.57^\circ$$

$$\theta_2 = \tan^{-1} \frac{2}{1} = 63.43^\circ$$

$$\theta_3 = \tan^{-1} \frac{1}{2} = 26.57^\circ$$

Now resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components.

$$+\rightarrow \Sigma H = 300 \cos 26.57^\circ - 200 \cos 63.43^\circ - 500 \cos 26.57^\circ$$

$$\therefore \Sigma H = -268.33 \text{ N}$$

(here '-' sign indicate that ΣH is acting toward negative x direction)

$$+\uparrow \Sigma V = 300 \sin 26.57^\circ + 200 \sin 63.43^\circ + 500 \sin 26.57^\circ$$

$$\therefore \Sigma V = 536.71 \text{ N}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(268.33)^2 + (536.71)^2}$$

$$\therefore R = 600 \text{ N}$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{536.71}{268.33} \quad \therefore \theta = 63.43^\circ \quad (\text{Ans.})$$

Example 3. Determine resultant of the force system shown in the figure Ex. 3.

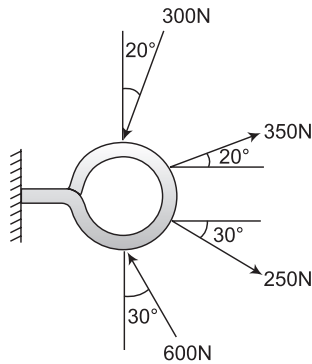


Fig. Ex. 3

Solution: Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components.

$$\begin{aligned} +\rightarrow \Sigma H &= -300 \sin 20^\circ + 350 \cos 20^\circ + 250 \cos 30^\circ - 600 \sin 30^\circ \\ &= 142.79 \text{ N} \end{aligned}$$

$$\begin{aligned} +\uparrow \Sigma V &= -300 \cos 20^\circ + 350 \sin 20^\circ - 250 \sin 30^\circ + 600 \cos 30^\circ \\ &= 232.41 \text{ N} \end{aligned}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(142.79)^2 + (232.41)^2}$$

$$\therefore R = 272.77 \text{ N} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{232.41}{142.79} \quad \therefore \theta = 58.43^\circ \quad (\text{Ans.})$$

Example 4. The following forces act at a point: 30 N towards East, 20 N towards North, 35 N towards North West, 25 N inclines at 40° towards South of West, 40 N towards South. Determine the magnitude of the resultant force and its direction from *East*.

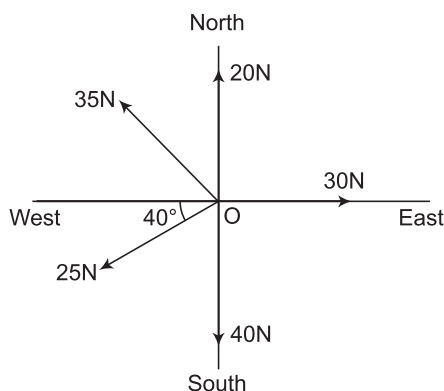


Fig. Ex. 4

Solution: The system of given forces is shown in *figure Ex. 4*.

Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components:

$$+\rightarrow \Sigma H = 30 - 35 \cos 45^\circ - 25 \cos 40^\circ$$

$= -13.90$ N (here ‘ $-$ ’ sign indicate that ΣH is acting toward negative x direction i.e. towards West)

$$+\uparrow \Sigma V = 20 + 35 \sin 45^\circ - 25 \sin 40^\circ - 40$$

$= -11.32$ N (here ‘ $-$ ’ sign indicate that ΣV is acting toward negative y direction i.e. towards South)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(13.90)^2 + (11.32)^2}$$

$$\therefore R = 17.92 \text{ N} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{11.32}{13.90}$$

$$\therefore \theta = 39.15^\circ$$

Now angle of resultant with East = $180^\circ + 39.15^\circ = 219.15^\circ$ (Ans.)

Example 5. Three forces act as shown in *figure Ex. 5*. Determine the magnitude of the force P if the resultant is along x-axis. Also determine magnitude of resultant.

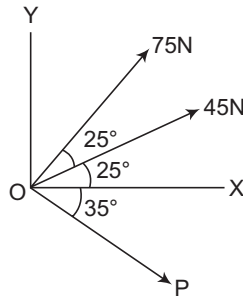


Fig. Ex. 5

Solution: As resultant is acting along x-axis, its vertical component become zero i.e. $\Sigma V = 0$.

$\therefore$ Resolving the forces along y-axes, and then equating sum of vertical components to zero,

$$\therefore 75 \sin 50^\circ + 45 \sin 25^\circ - P \sin 35^\circ = 0$$

$$\therefore P = 133.32 \text{ N (Ans.)}$$

Now as resultant is acting along x-axis, its magnitude is equal to ΣH

$$R = \Sigma H = 75 \cos 50^\circ + 45 \cos 25^\circ + 133.32 \cos 35^\circ$$

$$\therefore R = 198.20 \text{ N (Ans.)}$$

Example 6. Three forces are acting on a bracket as shown in *figure Ex. 6*. Determine the magnitude and direction of resultant force.

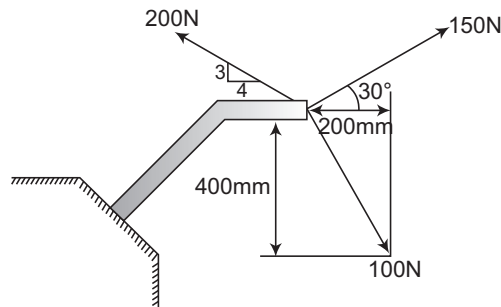


Fig. Ex. 6

Solution: Consider θ_1 , and θ_2 are the angles of forces 100 N, and 200 N respectively from horizontal.

$$\text{Now} \quad \theta_1 = \tan^{-1} \frac{400}{200} = 63.43^\circ$$

$$\theta_2 = \tan^{-1} \frac{3}{4} = 36.86^\circ$$

Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components:

$$\begin{aligned} \rightarrow \Sigma H &= 100 \cos 63.46^\circ + 150 \cos 30^\circ - 200 \cos 36.86^\circ \\ &= 14.56 \text{ N} \end{aligned}$$

$$\begin{aligned} +\uparrow \Sigma V &= -100 \sin 63.43^\circ + 150 \sin 30^\circ + 200 \sin 36.86^\circ \\ &= 105.53 \text{ N} \end{aligned}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(14.56)^2 + (105.53)^2}$$

$$\therefore R = 106.52 \text{ N} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{105.53}{14.56} \quad \therefore \theta = 82.14^\circ \quad (\text{Ans.})$$

Example 7. Four concurrent forces acts on the centre of mass during landing of airplane as shown in *figure Ex. 7*. Determine the magnitude of the resultant force and its direction from x -axis

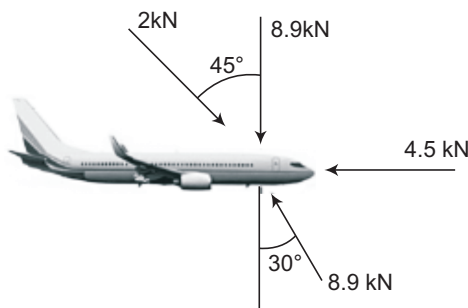


Fig. Ex. 7

Solution: Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components.

$$\begin{aligned} \rightarrow \Sigma H &= 2 \sin 45^\circ - 4.5 - 8.9 \sin 30^\circ \end{aligned}$$

$= -7.53 \text{ kN}$ (here '-' sign indicate that ΣH is acting toward negative x direction)

$$+\uparrow \Sigma V = -2 \cos 45^\circ - 8.9 + 8.9 \cos 30^\circ$$

$= -2.60 \text{ kN}$ (here '-' sign indicate that ΣV is acting toward negative y direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(7.53)^2 + (2.60)^2}$$

$$\therefore R = 7.97 \text{ kN} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{2.60}{7.53}$$

$$\therefore \theta = 19^\circ \quad (\text{Ans.})$$

Example 8. Bracket AB is supported by cable BC as shown in figure Ex. 8. If the tension in cable BC is 150 N , determine magnitude of the resultant of the three forces and direction of resultant from x -axis.

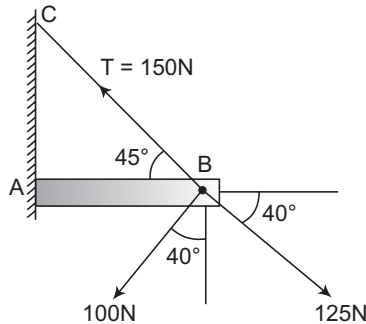


Fig. Ex. 8

Solution: Resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components:

$$+\rightarrow \Sigma H = -150 \cos 45^\circ - 100 \sin 40^\circ + 125 \cos 40^\circ$$

$= -74.58 \text{ N}$ (here '-' sign indicate that ΣH is acting toward negative x direction)

$$+\uparrow \Sigma V = 150 \sin 45^\circ - 100 \cos 40^\circ - 125 \sin 40^\circ$$

$= -50.88 \text{ N}$ (here '-' sign indicate that ΣV is acting toward negative y direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(74.58)^2 + (50.88)^2}$$

$$\therefore R = 90.28 \text{ N} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{50.88}{74.58} \therefore \theta = 34.30^\circ \quad (\text{Ans.})$$

Example 9. Three forces are acting on a hook as shown in *figure Ex. 9*. If the magnitude of the resultant force is 400 N and acting along positive y -axis, determine the magnitude of force P and the angle α .

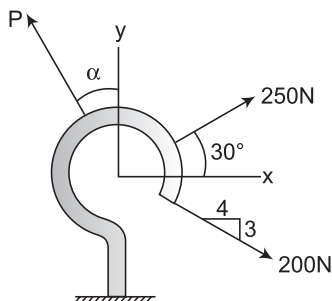


Fig. Ex. 9

Solution: Consider θ is the angle of 200 N force with x axis,

$$\text{Now} \quad \theta = \tan^{-1} \frac{3}{4} = 36.86^\circ$$

As resultant is acting along y -axis, its horizontal component become zero i.e. $\Sigma H = 0$.

$\therefore$ Resolving the forces along x -axes, and then equating sum of horizontal components to zero,

$$\therefore -P \sin \alpha + 250 \cos 30^\circ + 200 \cos 36.86^\circ = 0$$

$$\therefore P \sin \alpha = 376.52 \quad (\text{Eq. 1})$$

Now resultant is acting along y axis, its magnitude is equal to ΣV

$$\therefore 400 = \Sigma V$$

$$\therefore 400 = P \cos \alpha + 250 \sin 30^\circ - 200 \sin 36.86^\circ$$

$$P \cos \alpha = 394.98 \quad (\text{Eq. 2})$$

Solving Eq. 1 and Eq. 2,

$$\therefore \alpha = 43.62^\circ \quad (\text{Ans.})$$

$$\text{Now from (Eq. 1), } P \sin 43.63^\circ = 376.52 \quad P = 545.68 \text{ N} \quad (\text{Ans.})$$

Example 10. Three forces are acting on a bracket as shown in *figure Ex. 10*. If the magnitude of the resultant force is to be 250 N directed along positive u axis, determine force P and angle α .

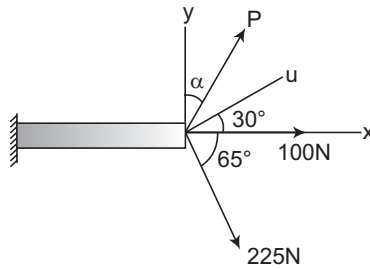


Fig. Ex. 10

Solution: Here find ΣH and equate with horizontal component of the resultant force,

$$\therefore \quad \rightarrow \Sigma H = 250 \cos 30^\circ$$

$$\therefore P \sin \alpha + 100 + 225 \cos 65^\circ = 250 \cos 30^\circ$$

$$\therefore P \sin \alpha = 21.41 \quad (\text{Eq.1})$$

Now find ΣV and equate with vertical component of the resultant force,

$$\therefore \quad +\uparrow \Sigma V = 250 \sin 30^\circ$$

$$\therefore P \cos \alpha - 225 \sin 65^\circ = 250 \sin 30^\circ$$

$$\therefore P \cos \alpha = 328.91 \quad (\text{Eq.2})$$

Solving Eq. 1 and Eq. 2,

$$\therefore \quad \alpha = 3.72^\circ \quad (\text{Ans.})$$

$$\text{Now from (Eq. 1), } P \sin 3.72^\circ = 21.41 \quad P = 330 \text{ N} \quad (\text{Ans.})$$

Solved Examples Based on Moment & Non-Concurrent Forces

Example 1. Determine the moment of the 100 N force about point O as shown in the figure Ex. 1.

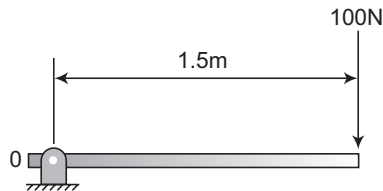


Fig. Ex. 1

Solution: Since the perpendicular distance d from the force to the point O is 1.5 m, we use

$$M_O = P \times d$$

Treating anticlockwise moment as positive

$$M_O = -100 \times 1.5$$

(Negative sign is used from our assumed sign convention of moment, as 100 N force tends to rotate the object in clockwise direction)

$$\therefore \quad M_O = -150 \text{ N.m}$$

$$M_O = 150 \text{ N.m (Clockwise) (Ans.)}$$

Example 2. Determine the moment of the 100 N force about point O as shown in the figure Ex. 2

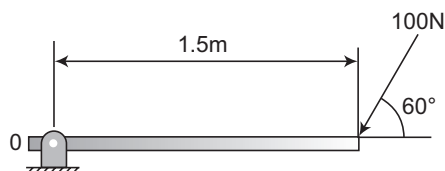


Fig. Ex. 2

Solution: Here first determine perpendicular distance from line of action of 100 N force to point O as shown in figure Ex. 2(a).

$$M_O = P \times d$$

$$\therefore \quad M_O = -100 \times 1.5 \sin 60^\circ$$

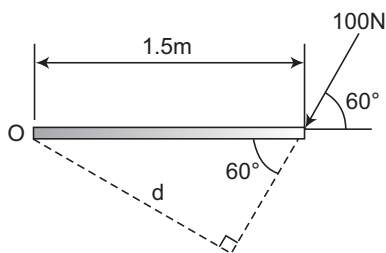


Fig. Ex. 2(a)

$$M_O = -129.9 \text{ N.m} = 129.9 \text{ N.m (clockwise)}$$

Alternate Approach: Many times it is convenient to resolve the given force into its components and then take algebraic sum of moment of these components about given point. In above example given 100 N force can be resolve into horizontal and vertical components as shown in figure Ex. 2(b).

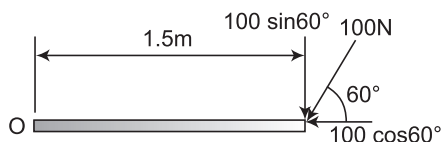


Fig. Ex. 2(b)

Treating anticlockwise moment as positive, sum of moment due to these components about point O is find out as below. (Note that, the line of action of horizontal component is passing through point O , so it produces zero moment)

$$\therefore \quad M_O = -100 \sin 60^\circ \times 1.5$$

$$M_O = -129.9 \text{ N.m} = 129.9 \text{ N.m (clockwise) (Ans.)}$$

Example 3. Determine the moment of the 100 N force about point O as shown in the figure Ex. 3.

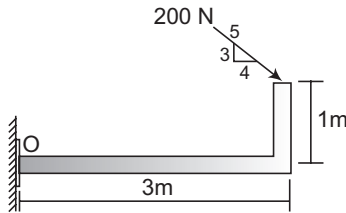


Fig. Ex. 3

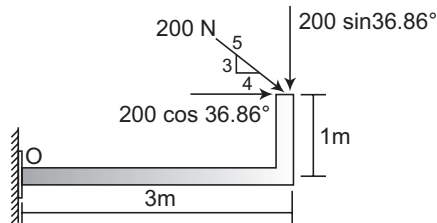


Fig. Ex. 3(a)

Solution: Here resolved 200 N force into horizontal and vertical components as shown in figure Ex. 3(a).

Treating anticlockwise moment as positive, sum of moment due to these components about point O is,

$$M_O = -200 \cos 36.86^\circ \times 1 - 200 \sin 36.86^\circ \times 3$$

$$\therefore M_O = -520 \text{ N.m} = 520 \text{ N.m (clockwise)} \quad (\text{Ans.})$$

Example 4. Three forces are acting in a plane as shown in figure Ex. 4. Determine moment of each forces about point A and B .

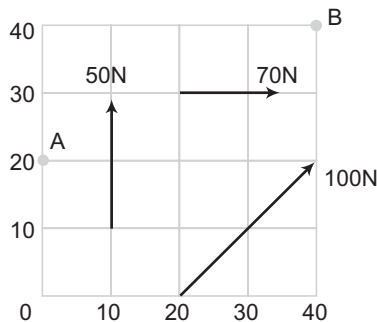


Fig. Ex. 4

Solution: Treating anticlockwise moment as positive, moment of 50 N force about point A and B are,

$$M_A = 50 \times 10 = 500 \text{ N.mm (anticlockwise) (Ans.)}$$

$$M_B = -50 \times 30 = -1500 \text{ N.mm} = 1500 \text{ N.mm (clockwise) (Ans.)}$$

Moment of 70 N force about point A and B are,

$$M_A = -70 \times 10 = -700 \text{ N.mm} = 700 \text{ N.mm (clockwise) (Ans.)}$$

$$M_B = 70 \times 10 = 700 \text{ N.mm (anticlockwise) (Ans.)}$$

Now resolved 100 N force into horizontal and vertical components and then find out moment due to these components about point A and B as below,

$$\text{Horizontal component of 100 N force} = 100 \cos 45^\circ = 70.71 \text{ N}$$

$$\text{Vertical component of 100 N force} = 100 \sin 45^\circ = 70.71 \text{ N}$$

$$M_A = (70.71 \times 20) + (70.71 \times 20) = 2828.4 \text{ N.mm (anticlockwise) (Ans.)}$$

$$M_B = (70.71 \times 40) - (70.71 \times 20) = 1414.2 \text{ N.mm (anticlockwise) (Ans.)}$$

Example 5. A bracket is subjected to two forces as shown in *figure Ex. 5*. Determine (a) moment of 160 N force about point A , (b) moment of 210 N force about point A and B .

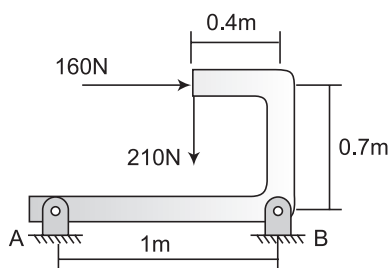


Fig. Ex. 5

Solution: Treating anticlockwise moment as positive, moment of 160 N force about point A is,

$$M_A = -160 \times 0.7 = -112 \text{ N.m} = 112 \text{ N.m (clockwise) (Ans.)}$$

Moment of 210 N force about point A is,

$$M_A = -210 \times 0.6 = -126 \text{ N.m} = 126 \text{ N.m (clockwise) (Ans.)}$$

$$M_B = 210 \times 0.4 = 84 \text{ N.m (anticlockwise) (Ans.)}$$

Example 6. Force P is acting in a plane as shown in *figure Ex. 6*. If the magnitude of the moment due to the force P about O is 300 N-m, determine magnitude of force P .

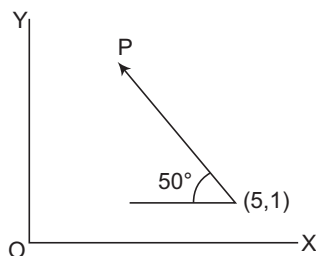


Fig. Ex. 6

Solution: Here first find out horizontal and vertical components of force P as shown in figure Ex. 6(a).

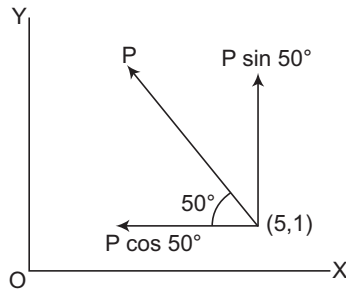


Fig. Ex. 6(a)

Treating anticlockwise moment as positive, sum of moment due to these components about point O is,

$$M_O = (P \sin 50^\circ \times 5) + (P \cos 50^\circ \times 1)$$

Now it is given that moment due to the force P about O is 300 N-m

$$\therefore 300 = (P \sin 50^\circ \times 5) + (P \cos 50^\circ \times 1)$$

$$\therefore 300 = P [(0.7660 \times 5) + (0.6427 \times 1)]$$

$$\therefore P = 67 \text{ N (Ans.)}$$

Example 7. Two equal forces of 150 N are applied as shown in figure Ex. 7. If the length of the bar AB is 500 mm and radius of pulley is 100 mm, determine sum of the moments of the forces (a) about point A and (b) about point B .

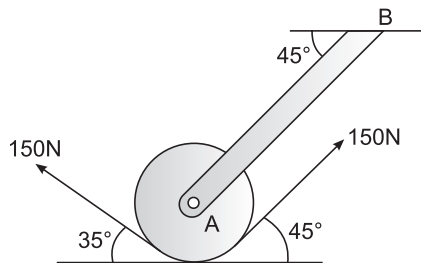


Fig. Ex. 7

Solution: Treating anticlockwise moment as positive, sum of moment due to these forces about point A is,

$$\Sigma M_A = (150 \times 100) - (150 \times 100)$$

$$\therefore M_A = 0 \text{ (Ans.)}$$

Now sum of moment about B is

$$\Sigma M_B = (150 \times 100) - (150 \cos 35^\circ) (500 \sin 45^\circ + 100 \cos 35^\circ) - (150 \sin 35^\circ) (500 \cos 45^\circ + 100 \sin 35^\circ)$$

$$\therefore M_B = -73860.57 \text{ N.mm} = -73.86 \text{ N.m} = 73.86 \text{ N.m (clockwise) (Ans.)}$$

Example 8. A 200 N force is acting on a bracket as shown in *figure Ex. 8*. Determine the value of θ for which the moment about O is (a) zero and (b) a maximum.

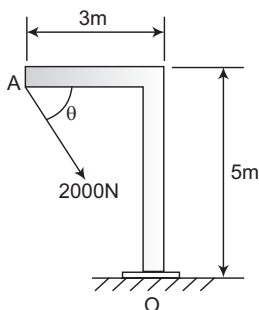


Fig. Ex. 8

Solution: For moment about O to be zero, the line of action of the given force must pass through point O as shown in *figure Ex. 8 (a)*.

$$\therefore \theta = \tan^{-1} \frac{5}{3} = 59^\circ \quad (\text{Ans.})$$

Now for moment about O to be a maximum, the line of action of the given force must be perpendicular to distance OA as shown in *figure Ex. 8 (b)*.

$$\therefore \theta = \left(\tan^{-1} \frac{5}{3} \right) + (90^\circ) = 149^\circ \quad (\text{Ans.})$$

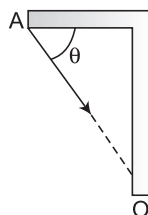


Fig. Ex. 8(a)

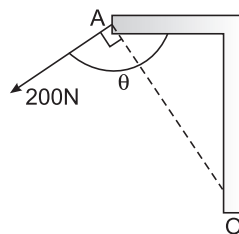


Fig. Ex. 8(b)

Example 9. Two forces acting on a bracket as shown in *figure Ex. 9*. Determine the sum of moment of these two forces about point O .

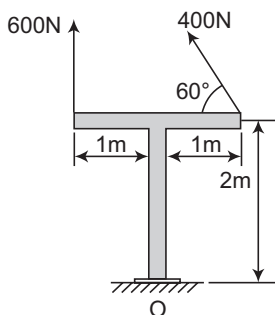


Fig. Ex. 9

Solution: Treating anticlockwise moment as positive, sum of moment due to these forces about point O is,

$$\Sigma M_O = -(600 \times 1) + (400 \sin 60^\circ \times 1) + (400 \cos 60^\circ \times 2)$$

$\therefore$

$$M_O = 146.41 \text{ (anticlockwise)} \quad (\text{Ans.})$$

Example 10. Cable AB and AC are used to support a pole as shown in *figure Ex. 10*. If the tension in cable AC is 250 N and the sum of the moments about O by the two cables is zero, determine the tension in cable AB .

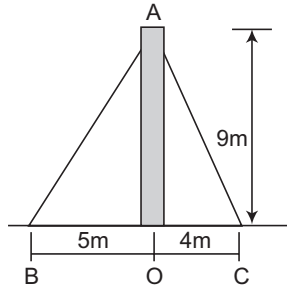


Fig. Ex. 10

Solution: Let θ_1 and θ_2 are the angles of cable AB and AC with horizontal respectively.

$$\theta_1 = \tan^{-1} \frac{9}{5} = 61^\circ \text{ and } \theta_2 = \tan^{-1} \frac{9}{4} = 66^\circ$$

Now find out components of the cable tensions at point A .

As sum of the moments about point O is zero, $\Sigma M_O = 0$

Treating anticlockwise moment as positive and T be the tension in cable AB ,

$$T \cos 61^\circ \times 9 - 250 \cos 66^\circ \times 9 = 0$$

$\therefore$

$$T = 209.74 \text{ N} \quad (\text{Ans.})$$

Example 11. Square $ABCD$ is subjected to four forces and clockwise moment as shown in *figure Ex. 11*. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point A .

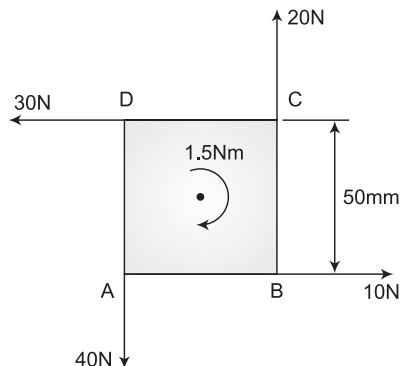


Fig. Ex. 11

Solution: Here first resolve the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$\rightarrow \Sigma H = 10 - 30 = -20 \text{ N}$ (here '-' sign indicate that ΣH is acting toward negative x Direction)

$+\uparrow \Sigma V = 20 - 40 = -20 \text{ N}$ (here '-' sign indicate that ΣV is acting toward negative y Direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(20)^2 + (20)^2}$$

$$\therefore R = 28.28 \text{ N (Ans.)}$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{20}{20} \quad \therefore \theta = 45^\circ \quad (\text{Ans.})$$

Now position of resultant force is determine by applying Varignon's theorem about point A . Let x be the perpendicular distance between line of action of the resultant force and point A .

$\therefore Rx = \Sigma M_A$, treating anticlockwise moment as positive

$\therefore 28.28 x = 30 \times 0.05 + 20 \times 0.05 - 1.5$ (note that the moment of 10 N and 40 N forces about point A is zero, as the line of action of these forces are passing through point A)

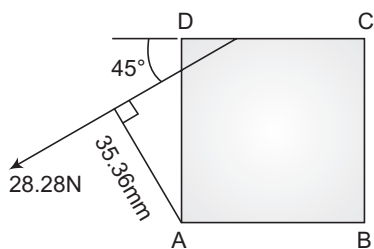


Fig. Ex. 11(a)

$\therefore x = 0.0353 \text{ m} = 35.36 \text{ mm}$ from point A , refer figure Ex. 11(a) (Ans.)

Example 12. A rectangular block $ABCD$ is subjected to four forces as shown in figure Ex. 12. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point A .

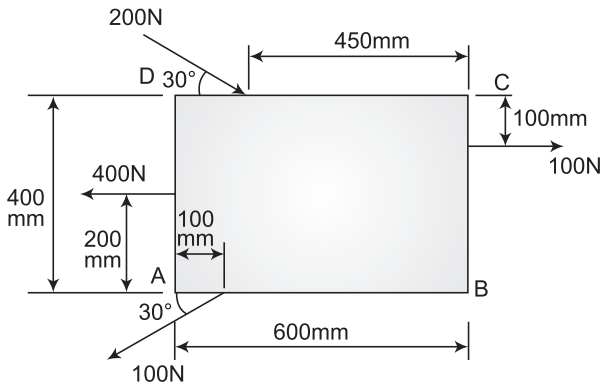


Fig. Ex. 12

Solution: Here first resolve the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$$+\rightarrow \Sigma H = 100 - 400 - 100 \cos 30^\circ + 200 \cos 30^\circ = -213.39 \text{ N}$$

(here '-' sign indicate that ΣH is acting toward negative x Direction)

$$+\uparrow \Sigma V = -200 \sin 30^\circ - 100 \sin 30^\circ = -150 \text{ N}$$

(here '-' sign indicate that ΣV is acting toward negative y Direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(213.39)^2 + (150)^2}$$

$$\therefore R = 260.83 \text{ N (Ans.)}$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{150}{213.39} \quad \therefore \theta = 35.10^\circ \text{ (Ans.)}$$

Now position of resultant force is determine by applying Varignon's theorem about point A . Let x be the perpendicular distance between line of action of the resultant force and point A .

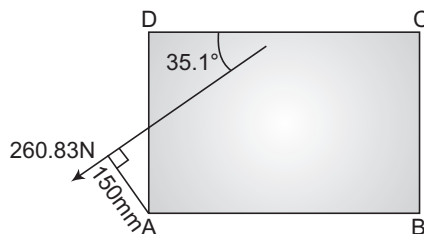


Fig. Ex. 12(a)

$$\begin{aligned}
 \therefore Rx &= \Sigma M_A, \text{ treating anticlockwise moment as positive} \\
 \therefore 260.83 x &= (400 \times 200) - (100 \times 300) - (100 \sin 30^\circ \times 100) \\
 &\quad - (200 \cos 30^\circ \times 400) - (200 \sin 30^\circ \times 150) \\
 \therefore x &= 0.150 \text{ m} = 150 \text{ mm from point } A, \text{ refer figure Ex. 12(a)} \quad (\text{Ans.})
 \end{aligned}$$

Example 13. Three forces are acting on member AB as shown in figure Ex. 13. Determine (a) magnitude and direction of the resultant force, (b) point where line of action of resultant intersect member AB .

Solution: Here first resolve the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$$\rightarrow \Sigma H = -25 \cos 45^\circ + 20 \sin 25^\circ = -9.22 \text{ N}$$

(here ‘-’ sign indicate that ΣH is acting toward negative x Direction)

$$+\uparrow \Sigma V = -30 - 25 \sin 45^\circ - 20 \cos 25^\circ = -65.80 \text{ N}$$

(here ‘-’ sign indicate that ΣV is acting toward negative y Direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(9.22)^2 + (65.80)^2}$$

$$\therefore R = 66.44 \text{ N} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{65.80}{9.22} \quad \therefore \theta = 82^\circ \quad (\text{Ans.})$$

Now position of resultant force is determine by applying Varignon’s theorem about point A . Let x be the perpendicular distance between line of action of the resultant force and point A .

$$\begin{aligned}
 \therefore Rx &= \Sigma M_A, \text{ treating anticlockwise moment as positive} \\
 \therefore 66.44 x &= (-30 \times 1) - (25 \sin 45^\circ \times 2) - (20 \cos 25^\circ \times 4) \\
 \therefore x &= 2.07 \text{ m from point } A, \text{ refer figure Ex. 13(a)}
 \end{aligned}$$

Now point where line of action of resultant intersect member AB from A as shown in figure Ex. 13(a) is determine by again applying Varignon’s theorem as below,

$$\Sigma V \times \text{horizontal distance } AC = \Sigma M_A$$

$$65.80 \times \text{distance } AC = (-30 \times 1) - (25 \sin 45^\circ \times 2) - (20 \cos 25^\circ \times 4)$$

$$\therefore \text{distance } AC = 2.09 \text{ m} \quad (\text{Ans.})$$

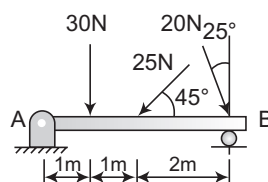


Fig. Ex. 13

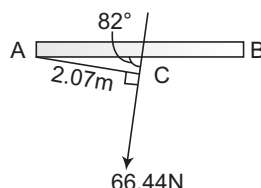


Fig. Ex. 13(a)

Example 14. A bracket is subjected to four forces as shown in *figure Ex. 14*.

Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point *O*.

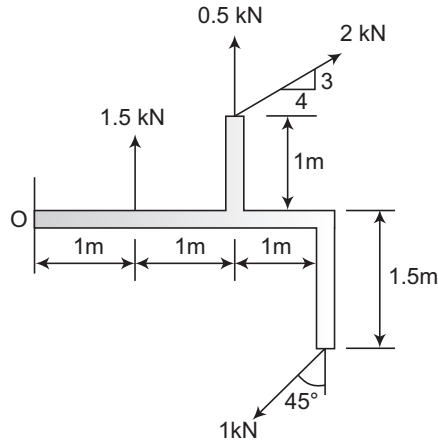


Fig. Ex. 14

Solution: Let θ be the angle of 2 kN force with horizontal.

$$\theta = \tan^{-1} \frac{3}{4} = 36.86^\circ$$

Now resolving the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$$+\rightarrow \Sigma H = 2 \cos 36.86^\circ - 1 \sin 45^\circ = 0.8931 \text{ kN}$$

$$+\uparrow \Sigma V = 1.5 + 0.5 + 2 \sin 36.86^\circ - 1 \cos 45^\circ = 2.49 \text{ kN}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(0.8931)^2 + (2.49)^2}$$

$$\therefore R = 2.65 \text{ kN} \quad (\text{Ans.})$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{2.49}{0.8931} \quad \therefore \theta = 70.26^\circ \quad (\text{Ans.})$$

Now position of resultant force is determine by applying Varignon's theorem about point *O*. Let x be the perpendicular distance between line of action of the resultant force and point *O*.

$$\therefore R x = \Sigma M_A, \text{ treating anticlockwise moment as positive}$$

$$\therefore x = 0.0442 \text{ m from point } O \quad \textbf{(Ans.)}$$

Solved Examples Based on Parallel Forces

Example 1. Determine the magnitude of the resultant of parallel force system shown in *figure Ex. 1*. Also determine the position of the resultant force from point *O*.

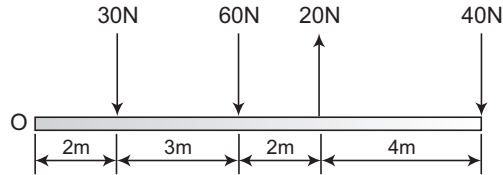


Fig. Ex. 1

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = +\uparrow \Sigma V = -30 - 60 + 20 - 40$$

$$\therefore R = -110 \text{ N} = 110 \text{ N downward (Ans.)}$$

Now position of resultant is finding out by applying Varignon's theorem at point *O*.

$\therefore$ Moment due to resultant *R* about point *O* = Sum of moment due to all forces about point *O*

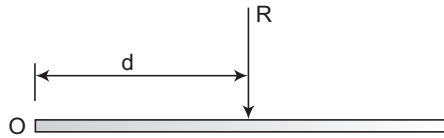


Fig. Ex. 1(a)

$$\therefore -R \times d = -(30 \times 2) - (60 \times 5) + (20 \times 7) - (40 \times 11)$$

(Treating anticlockwise moment as positive)

$$-110 \times d = -660$$

$$\therefore d = 6 \text{ m Refer figure Ex. 1(a)}$$

Example 2. A pole is subjected to parallel forces as shown in *figure Ex. 2*. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point *O*.

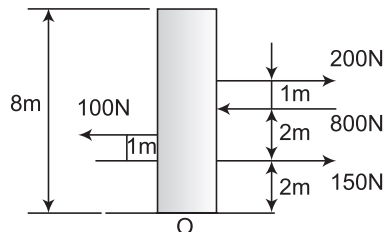


Fig. Ex. 2

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = \overset{+}{\rightarrow} \Sigma H = 200 - 800 - 100 + 150$$

$$\therefore R = -550 \text{ N} = 550 \text{ N acting towards left (Ans.)}$$

Now position of resultant is finding out by applying Varignon's theorem at point O .

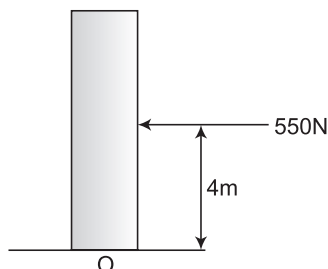


Fig. Ex. 2(a)

$\therefore$ Moment due to resultant R about point O = Sum of moment due to all forces about point O

$$\therefore 550 \times d = -150 \times 2 + 100 \times 3 + 800 \times 4 - 200 \times 5$$

(Treating anticlockwise moment as positive)

$$\therefore d = 4 \text{ m from } O \text{ Refer figure Ex.2(a) (Ans.)}$$

Example 3. Three parallel forces are acting on the lever as shown in figure Ex.3. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point O .

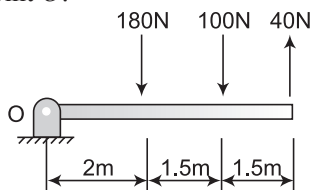


Fig. Ex. 3

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = +\uparrow \Sigma V = -180 - 100 + 40$$

$$\therefore R = -240 \text{ N} = 240 \text{ N downward (Ans.)}$$

Now position of resultant is finding out by applying Varignon's theorem at point O .

$\therefore$ Moment due to resultant R about point O
= Sum of moment due to all forces about point O

$$\therefore -R \times d = -(180 \times 2) - (100 \times 3.5) + (40 \times 5)$$

(Treating anticlockwise moment as positive)

$$-240 \times d = -510$$

$$\therefore d = 2.125 \text{ m Refer figure Ex.3(a) (Ans.)}$$

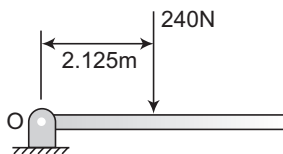


Fig. Ex. 3(a)

Example 4. Four parallel forces are acting on the lever as shown in *figure Ex.4*. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point O .

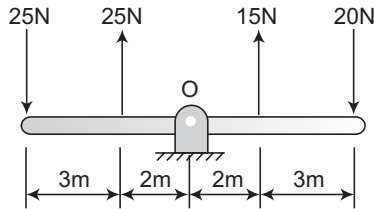


Fig. Ex. 4

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = +\uparrow \Sigma V = -25 + 25 + 15 - 20$$

$$\therefore R = -5 \text{ N} = 5 \text{ N downward (Ans.)}$$

Now position of resultant is finding out by applying Varignon's theorem at point O .

$\therefore$ Moment due to resultant R about point O = Sum of moment due to all forces about point O

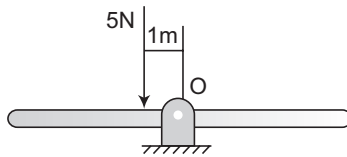


Fig. Ex. 4(a)

$$\therefore R \times d = (25 \times 5) - (25 \times 2) + (15 \times 2) - (20 \times 5)$$

(Treating anticlockwise moment as positive)

$$5 \times d = 5$$

$$\therefore d = 1 \text{ m to the left of point } O \text{ Refer figure Ex.4(a) (Ans.)}$$

Example 5. Four parallel forces are acting on lever as shown in *figure Ex.5*. Determine magnitude of P_1 and P_2 so that the four forces produce downward resultant of 150 N acting at 4 m from O

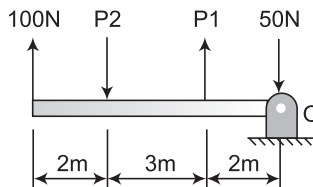


Fig. Ex. 5

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = +\uparrow \Sigma V = 100 - P_2 + P_1 - 50$$

Now it is given that magnitude of resultant force is 150 N

$$\therefore -150 = -50 + P_1 - P_2 + 100$$

$$\therefore P_1 - P_2 = -200 \quad \therefore P_1 = P_2 - 200 \quad (\text{Eq. 1})$$

Now by applying Varignon's theorem at point O ,

$\therefore$ Moment due to resultant R about point O = Sum of moment due to all forces about point O

$$\therefore 150 \times 4 = -2P_1 + 5P_2 - 700$$

(Treating anticlockwise moment as positive)

$$2P_1 - 5P_2 + 1300 = 0$$

$$2(P_2 - 200) - 5P_2 + 1300 = 0 \quad \text{Substituting value of } P_1 \text{ from (Eq. 1)}$$

$$2P_2 - 400 - 5P_2 + 1300 = 0$$

$$900 = 5P_2 - 2P_2$$

$$900 = 3P_2$$

$$\therefore P_2 = 300 \text{ N (Ans.)}$$

Substituting value of P_2 in (Eq. 1)

$$P_1 = 300 - 200 \quad P_1 = 100 \text{ N (Ans.)}$$

Example 6. Four parallel forces are acting on circle of diameter 2 m as shown in figure Ex. 6. Determine (a) magnitude and direction of the resultant force and (b) position of the resultant force from point O .

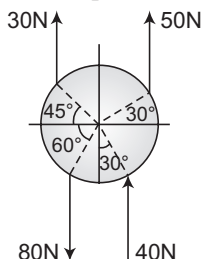


Fig. Ex. 6

Solution: Here magnitude of resultant is algebraic sum of all parallel forces.

$$R = +\uparrow \Sigma V = 50 + 30 + 40 - 80$$

$$\therefore R = 40 \text{ N upward (Ans.)}$$

Now position of resultant is finding out by applying Varignon's theorem at point O .

$\therefore$ Moment due to resultant R about point O = Sum of moment due to all forces about point O

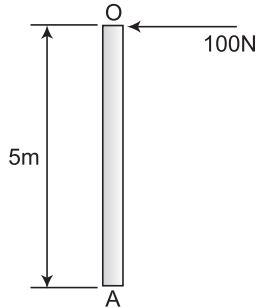
$$\therefore 40 \times d = (50 \times 1 \cos 30^\circ) - (30 \times 1 \cos 45^\circ) + (80 \times 1 \cos 60^\circ) + (40 \times 1 \sin 30^\circ) \quad (\text{Treating anticlockwise moment as positive})$$

$$40 \times d = 82$$

$$\therefore d = 2.05 \text{ m from point } O \quad (\text{Ans.})$$

Solved Examples Based on Force-Couple System and Equivalent System

Example 1. Replace 100 N force shown in *figure Ex. 1* by equivalent force couple at point *A*.

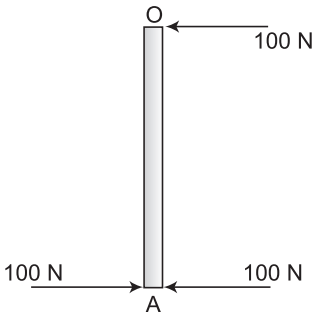
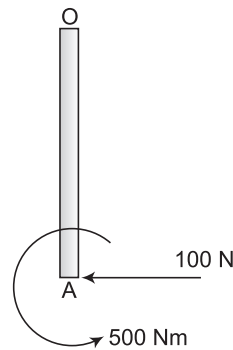
**Fig. Ex. 1**

Solution: Here transfer the force parallel to itself along with its moment at *A* as below:

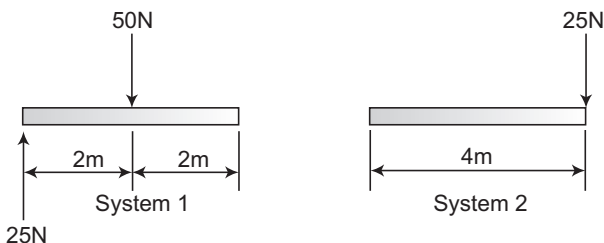
Introduce a system of 100 N forces at point *A* as shown in *figure Ex. 1(a)*

Now force 100 N acting at point *O* is replaced at point *A* by keeping 100 N force as it is and a couple moment as shown in *figure Ex. 1(b)*

$$M = 100 \times 5 = 500 \text{ N.m anticlockwise. (Ans.)}$$

**Fig. Ex. 1(a)****Fig. Ex. 1(b)**

Example 2. Two systems of forces are shown in *figure Ex. 2*. Are they equivalent?

**Fig. Ex. 2**

Solution: As we know that two systems are equivalent only when the sum of the forces and sum of the moments about an arbitrary point are equal.

Now check for sum of the forces of two systems.

$$+\uparrow \Sigma F_{\text{System 1}} = 25 - 50 = -25 \text{ N}$$

$$+\uparrow \Sigma F_{\text{System 2}} = -25 \text{ N}$$

Now check for sums of the moments of two systems about left end. Treating anticlockwise moment as positive

$$\Sigma M_{\text{System 1}} = -50 \times 2 = -100 \text{ N.m} = 100 \text{ N.m clockwise}$$

$$\Sigma M_{\text{System 2}} = -25 \times 4 = -100 \text{ N.m} = 100 \text{ N.m clockwise}$$

As sums of the forces of two systems are equal and sum of the moments of two systems about an arbitrary point are equal. Two systems are equivalent.

Example 3. Replace the 200 N force shown in *figure Ex. 3* by equivalent force couple at point A.

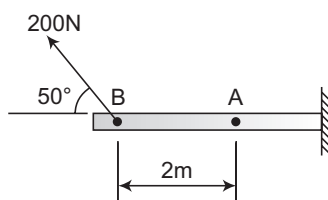


Fig. Ex. 3

Solution: Here transfer the force parallel to itself along with its moment at A as below:

Introduce a system of 200 N forces at point A as shown in *figure Ex. 3(a)*

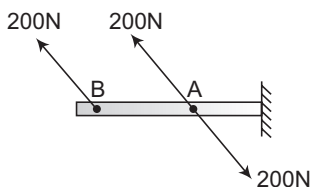


Fig. Ex. 3(a)

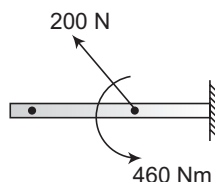


Fig. Ex. (b)

Now force 200 N acting at point B is replaced at point A by keeping 200 N force as it is and a couple moment as shown in *figure Ex. 3(b)*

$$M = 200 \sin 50^\circ \times 3 = 460 \text{ N.m clockwise. (Ans.)}$$

Example 4. The 50 N force is acting on lever as shown in *figure Ex. 4*. Replace the force by force and couple moment at point O that will have an equivalent effect.

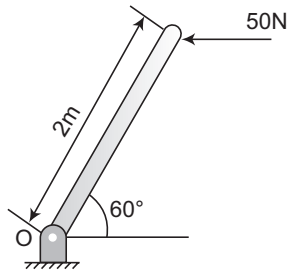


Fig. Ex. 4

Solution: Here transfer the force parallel to itself along with its moment at O as below:

Introduce a system of 50 N forces at point O as shown in figure Ex. 4(a)

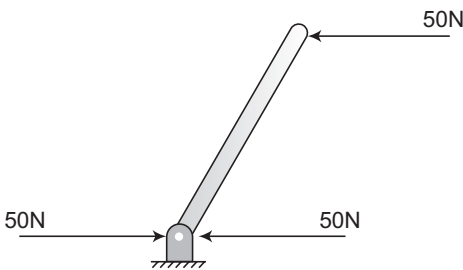


Fig. Ex. 4(a)

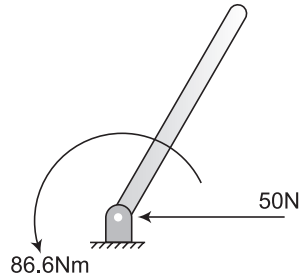


Fig. Ex. 4(b)

Now force 200 N acting at point A is replaced at point O by keeping 50 N force as it is and a couple moment as shown in figure Ex. 4(b)

$$M = 50 \times 2 \sin 60^\circ = 86.60 \text{ N.m anticlockwise. (Ans.)}$$

Example 5. A plate is subjected to two couples as shown in figure Ex.5. Determine the sum of the moments exerted on the plate by the two couples.

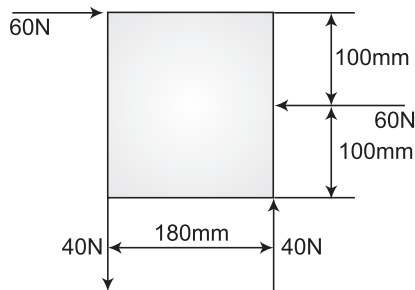


Fig. Ex. 5

Solution: Here, 60 N couple acts in clockwise direction and 40 N couple acts in anticlockwise direction. Now treating anticlockwise as positive,

$$\begin{aligned} \therefore \Sigma M &= -60 \times 100 + 40 \times 180 \\ &= -6000 + 7200 \end{aligned}$$

$$\therefore \Sigma M = 1200 \text{ N.mm anticlockwise (Ans.)}$$

Example 6. A plate is subjected to two couples as shown in *figure Ex.6*. Determine the magnitude of force P if the resultant couple moment is 300 N.m anticlockwise.

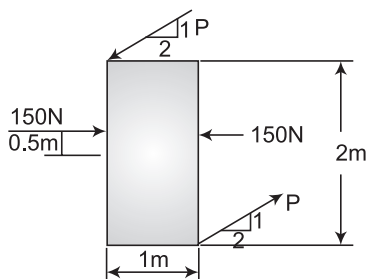


Fig. Ex. 6

Solution: Let θ be the angle of force P with horizontal.

$$\therefore \tan \theta = \frac{1}{2} \quad \therefore \theta = 26.57^\circ$$

Now resultant couple moment = sum of the moments exerted on the plate by the two couples.

Treating anticlockwise moment as positive,

$$\therefore 300 = (P \cos 26.57^\circ \times 2) + (P \sin 26.57^\circ \times 1) - (150 \times 0.5)$$

$$\therefore P = 168.16 \text{ N} \quad (\text{Ans.})$$

Example 7. A rectangular plate is subjected to a force – couple moment system as shown in *figure Ex.7*. Replace this system by a single force and determine its position from x -axis.

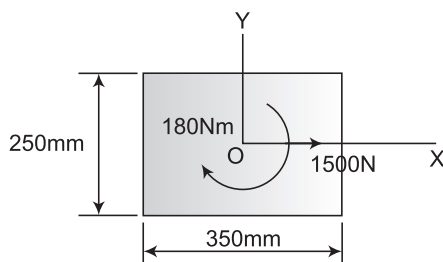


Fig. Ex. 7

Solution: Here first replace clockwise couple moment by two equal and opposite forces of magnitude 1500 N as shown in left hand part of *figure Ex.7(a)*. The 1500 N force acting towards left at point O and another 1500 N force acting towards right at point A, forms a couple. Here arm of couple is d and moment of couple is given by $M = P \times d$, where $M = 180 \text{ N.m}$ clockwise, $P = 1500 \text{ N}$ and d is unknown.

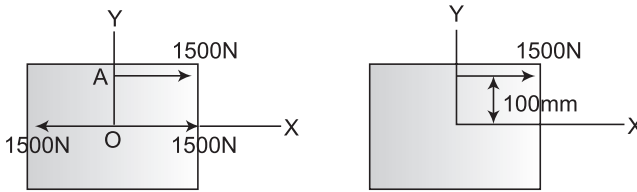


Fig. Ex. 7(a)

$\therefore d = \frac{M}{P} = \frac{150}{1500} = 0.1 \text{ m} = 100 \text{ mm}$ from x-axis as shown in the right hand part of figure Ex. 7(a). (Ans.)

Example 8. Replace the 100 N force actin at A on a square plate as shown in figure Ex. 8 by equivalent force and couple moment at point O.

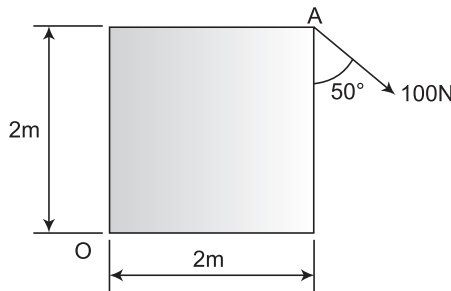


Fig. Ex. 8

Solution: Here transfer the force parallel to itself along with its moment at O as below:

Introduce 100 N forces at point O as shown in the left hand part of figure Ex. 8(a).

Now calculate the moment of a 100 N force about point O, (treating anticlockwise as positive)

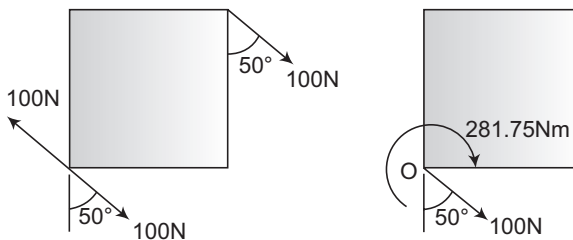


Fig. Ex. 8(a)

$$M = -100 \cos 50^\circ \times 2 - 100 \sin 50^\circ \times 2$$

$$M = -281.75 = 281.75 \text{ N.m clockwise. (Ans.)}$$

equivalent force and couple moment at point O is shown in the right hand part of figure Ex. 8(a).

Example 9. Two couples act on a pole as shown in *figure Ex. 9*. Determine the magnitude of P so that resultant couple moment is 300 N.m clockwise.

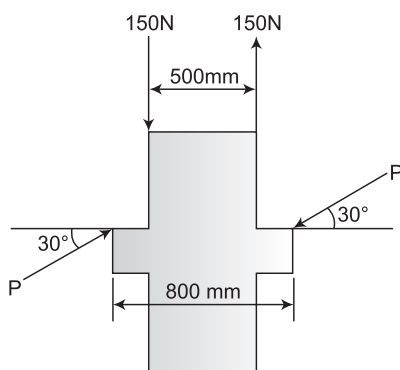


Fig. Ex. 9

Solution: The resultant couple moment is 300 N.m clockwise,

$$\therefore -300 = \text{sum of moments of two couples (treating anticlockwise as positive)}$$

$$\therefore -300 = 150 \times 500 - P \times 800 \sin 30^\circ \quad (\text{here } 800 \sin 30^\circ \text{ is the arm of couple } P)$$

$$\therefore P = 188.25 \text{ N} \quad (\text{Ans.})$$

Example 10. Replace the force system as shown in *figure Ex. 10* by resultant force and couple moment at point O that will have an equivalent effect.

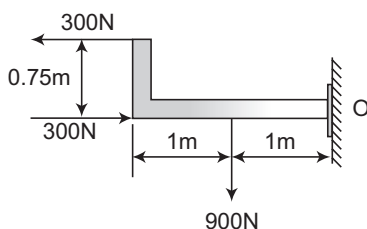


Fig. Ex. 10

Solution: Here first determine the resultant force as below,

$$+\rightarrow \Sigma H = -300 + 300 = 0$$

$$+\uparrow \Sigma V = -900 \text{ N (here '-' sign indicate that } \Sigma V \text{ is acting toward negative } y \text{ direction)}$$

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(0)^2 + (900)^2}$$

$$\therefore R = 900 \text{ N}$$

Now determine sum of moments of all forces about point O , (treating anticlockwise as positive)

$$\Sigma M_O = 300 \times 0.75 + 900 \times 1 = 1125 \text{ N.m anticlockwise}$$

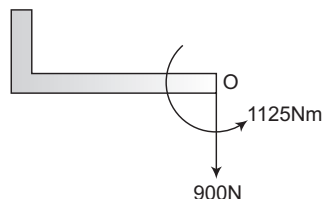


Fig. Ex. 10(a)

Resultant force and couple moment at point O of given system is shown in figure Ex.10(a).

Example 11. Replace the force system acting on a bracket as shown in figure Ex. 11 by resultant force and couple moment at point A that will have an equivalent effect.

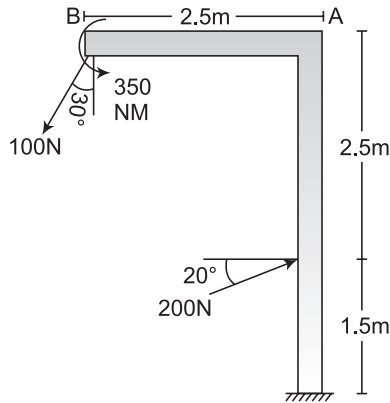


Fig. Ex. 11

Solution: Here first resolve the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$$+\rightarrow \Sigma H = 200 \cos 20^\circ - 100 \sin 30^\circ = 138 \text{ N}$$

$$+\uparrow \Sigma V = 200 \sin 20^\circ - 100 \cos 30^\circ = -18.19 \text{ N}$$

(here '-' sign indicate that ΣV is acting toward negative y direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(138)^2 + (18.19)^2}$$

$$\therefore R = 139.19 \text{ N}$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{18.19}{138} \quad \therefore \theta = 7.50^\circ$$

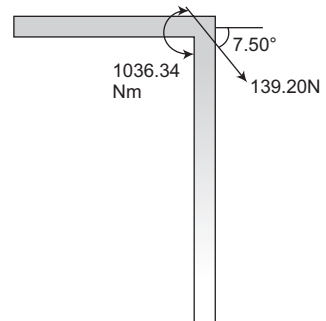


Fig. Ex. 11(a)

Now determine sum of moments of all forces about point A , (treating anticlockwise as positive)

$$\Sigma M_A = 350 + 100 \cos 30^\circ \times 2.5 + 200 \cos 20^\circ \times 2.5 = 1036.34 \text{ N.m anticlockwise}$$

Resultant force and couple moment at point A of given system is shown in figure Ex.11 (a).

Example 12. A square plate of 800 mm side is subjected to a couple and force as shown in *figure Ex. 12*. Replace the force system acting on plate with an equivalent force – couple system at O . (O is center of square plate)

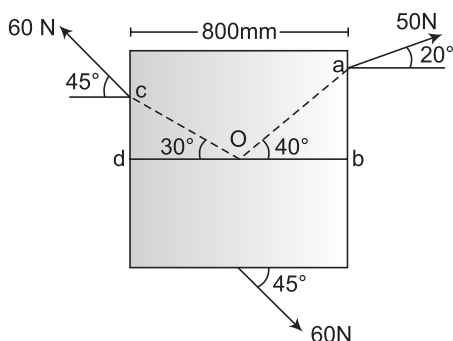


Fig. Ex. 12

Solution: It is identifying from given figure that two 60 N forces form a couple, therefore resultant force of the given force system is the remaining force i.e. 50 N.

$\therefore R = 50$ N and its inclination θ with horizontal axis is 20° .

For moment purpose first determine distance ab and cd . Refer *figure Ex. 12*.

In triangle oab

$$oa \cos 40^\circ = 400 \quad \therefore oa = 522.16 \text{ mm}$$

$$ab = 522.16 \sin 40^\circ \quad \therefore ab = 335.64 \text{ mm}$$

In triangle ocd

$$oc \cos 30^\circ = 400 \quad \therefore oc = 461.89 \text{ mm}$$

$$cd = 461.89 \sin 30^\circ \quad \therefore cd = 230.94 \text{ mm}$$

Now determine sum of moments of all forces about O , (treating anticlockwise as positive),

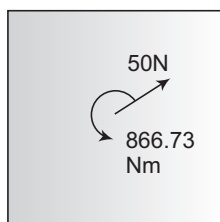


Fig. Ex. 12(a)

$$\begin{aligned} \Sigma M_O &= -(50 \cos 20^\circ \times 335.64) + (50 \sin 20^\circ \times 400) - (60 \sin 45^\circ \times 400) \\ &\quad + (60 \cos 45^\circ \times 230.90) + (60 \cos 45^\circ \times 400) \\ &= 866.73 \text{ N.m anticlockwise} \end{aligned}$$

Resultant force and couple moment at point O of given system is shown in *figure Ex. 12 (a)*.

Example 13. A bracket is subjected to three forces as shown in *figure Ex. 13*. Replace the force system with an equivalent force – couple system at B.

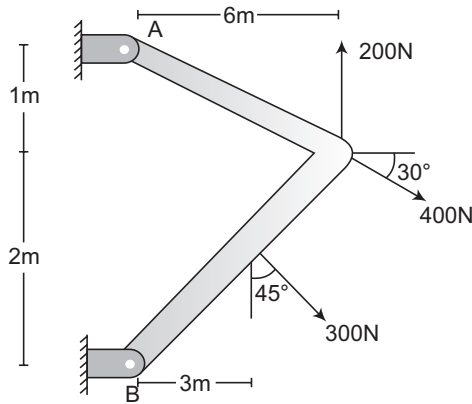


Fig. Ex. 13

Solution: Here first resolve the forces along x and y -axes, and then determine algebraic sum of horizontal components and vertical components,

$$\rightarrow \Sigma H = 400 \cos 30^\circ + 300 \sin 45^\circ = 558.54 \text{ N}$$

$$+\uparrow \Sigma V = 200 - 400 \sin 30^\circ - 300 \cos 45^\circ = -212.13 \text{ N}$$

(here ‘-’ sign indicate that ΣV is acting toward negative y direction)

Now we know that resultant R is calculated as,

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$$

$$\therefore R = \sqrt{(558.54)^2 + (212.13)^2}$$

$$\therefore R = 597.47 \text{ N}$$

Let θ be the angle of resultant R with x -axis, then

$$\tan \theta = \frac{\Sigma V}{\Sigma H}$$

$$\therefore \tan \theta = \frac{212.13}{558.54} \therefore \theta = 20.8^\circ$$

Now determine sum of moments of all forces about B, (treating anticlockwise as positive)

$$\begin{aligned} \Sigma M_B &= (200 \times 6) - (400 \cos 30^\circ \times 2) \\ &\quad - (400 \sin 30^\circ \times 6) - (300 \sin 45^\circ \times \left(2 - \frac{3}{6} \times 2\right)) - (300 \cos 45^\circ \times 3) \\ &= -1541.34 \text{ N.m} = 1541.34 \text{ N.m clockwise} \end{aligned}$$

Resultant force and couple moment at point A of given system is shown in *figure Ex. 13 (a)*.

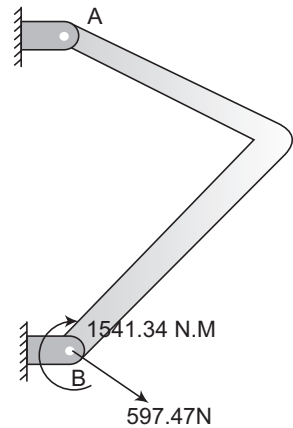


Fig. Ex. 13(a)

Example 14. Two systems of forces are shown in *figure Ex. 14*. Determine the value of R and P if these two systems are equivalent.

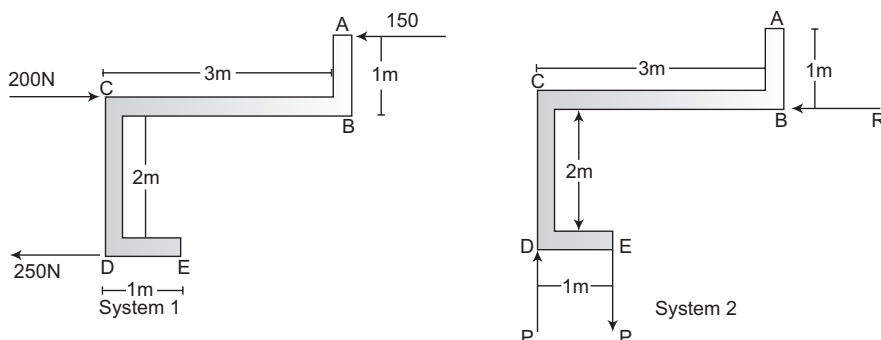


Fig. Ex. 14

Solution: As we know that two systems are equivalent only when the sum of the forces and sum of the moments about an arbitrary point are equal.

$$\Sigma H_{\text{System 1}} = \Sigma H_{\text{System 2}}$$

$$\therefore -150 + 200 - 250 = -R \quad \therefore R = 200 \text{ N (Ans.)}$$

Now determine sums of the moments of two systems about B and equate them. (Treating anticlockwise moment as positive)

$$\Sigma M_{\text{System 1B}} = \Sigma M_{\text{System 2B}}$$

$$150 \times 1 - 250 \times 2 = -P \times 1 \quad \therefore P = 350 \text{ N (Ans.)}$$

Example 15. Two systems of forces are shown in *figure Ex. 15*. Are they equivalent?

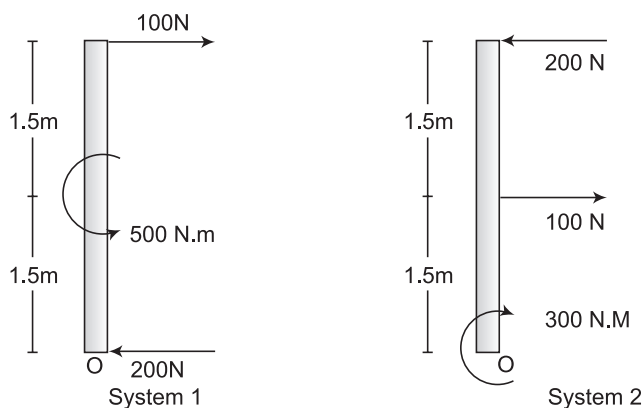


Fig. Ex. 15

Solution: As we know that two systems are equivalent only when the sum of the forces and sum of the moments about an arbitrary point are equal.

Now check for sum of the forces of two systems.

$$\rightarrow \Sigma F_{\text{System 1}} = 100 - 200 = -100 \text{ N}$$

$$\rightarrow \Sigma F_{\text{System 2}} = -200 + 100 = -100 \text{ N}$$

Now check for sums of the moments of two systems about O . (Treating anticlockwise moment as positive)

$$\Sigma M_{\text{System 1}} = -100 \times 3 + 500 = 200 \text{ N.m anticlockwise}$$

$$\Sigma M_{\text{System 2}} = 200 \times 3 - 100 \times 1.5 - 300 = 150 \text{ N.m anticlockwise}$$

As sums of the forces of two systems are equal but sum of the moments of two systems about an arbitrary point are not equal. Two systems are not equivalent.

Example 16. Two equivalent system of forces and moments act on the plate are shown in figure Ex. 16. Determine the force P and couple M .

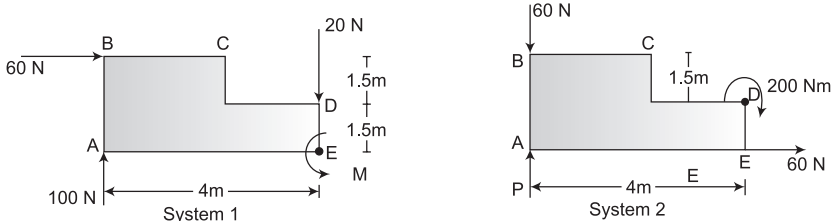


Fig. Ex. 16

Solution: As we know that two systems are equivalent only when the sum of the forces and sum of the moments about an arbitrary point are equal.

$$\rightarrow \Sigma H_{\text{System 1}} = \rightarrow \Sigma H_{\text{System 2}}$$

$$60 \text{ N} = 60 \text{ N}$$

$$+\uparrow \Sigma V_{\text{System 1}} = +\uparrow \Sigma V_{\text{System 2}}$$

$$\therefore 100 - 20 = P - 60 \quad \therefore P = 140 \text{ N (Ans.)}$$

Now determine sums of the moments of two systems about A and equate them. (Treating Anticlockwise moment as positive)

$$\rightarrow \Sigma M_{\text{System 1}_A} = \rightarrow \Sigma M_{\text{System 2}_A}$$

$$-60 \times 3 - 20 \times 4 + M = 200 \quad \therefore M = 460 \text{ N.m (Ans.)}$$

SUMMARY

- Force is the action of a body about another body.
 - ♦ Unit of the force is N, kN
 - ♦ $1 \text{ N} = (1 \text{ kg}) (1 \text{ m/s}^2) = 1 \text{ kg-m/s}^2$
- Magnitude, Direction, Sense and Point of application are characteristics of the force.
- When two or more forces act on a body they are called to form a system of forces.
 - ♦ Coplanar forces are those forces whose lines of action lie on the same plane.
 - ♦ Non-Coplanar forces are those forces whose lines of action do not lie on the same plane
- Resultant is a single force which produces the same effect as produced by number of forces when acting together.
- The resultant of two or more collinear forces is simply their algebraic sum.
- The resultant of two concurrent forces can be found out by means of the parallelogram law or the triangle law.

$$R = \sqrt{P_1^2 + P_2^2 + 2P_1P_2 \cos \theta} \quad \text{..... Magnitude of resultant}$$

$$\tan \alpha = \frac{P_2 \sin \theta}{P_1 + P_2 \cos \theta} \quad \text{..... Direction of resultant}$$

- The resultant of more than two concurrent forces can be found out by means of rectangular components method or the polygon law.

$$R = \sqrt{(\Sigma H)^2 + (\Sigma V)^2} \quad \text{..... Magnitude of resultant}$$

$$\tan \theta = \frac{\Sigma V}{\Sigma H} \quad \text{..... Direction of resultant}$$

- Resolution is the process of splitting up the given force into components, without changing its effect on the body.
- The rotational effect produced by force is known as moment of the force.

$$\text{Moment} = \text{Force} \times \text{Distance}$$

$$M = F \times d$$

- The location of the resultant of parallel force system is found out by means of Varignon's Theorem (Principle of Moment)
- Two equal, unlike parallel, non-collinear forces form a couple.
- Force can be transfer parallel to itself by means of force couple system.
- Two force systems that produce the same external effects on a rigid body are said to be equivalent.

PROBLEMS

1. Two forces of 400 N and 150 N are acting on a hook as shown in *figure Prob.1*. Determine magnitude resultant of these two forces.

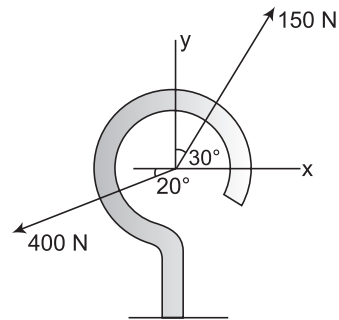


Fig. Prob. 1

2. Two forces of 375 N and P are acting on a hook as shown in *figure Prob.2*. If the magnitude of the resultant force is to be 200 N, directed along positive y -axis, determine the magnitude of force P .

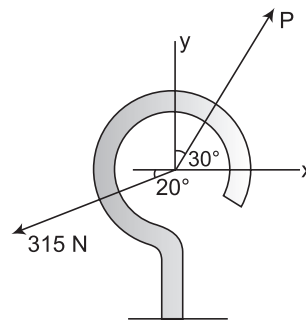


Fig. Prob. 2

3. Two forces of 150 N and 200 N are acting on a plate as shown in *figure Prob.3*. Determine magnitude and direction of resultant of these two forces.

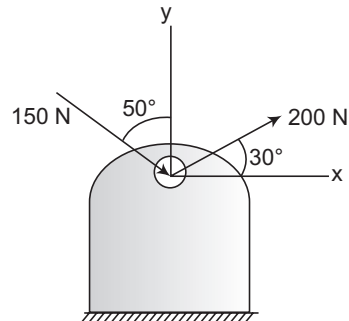


Fig. Prob. 3

4. Two forces of 160 N and 200 N are acting on a bracket as shown in *figure Prob.4*. Determine magnitude and direction of resultant of these two forces.

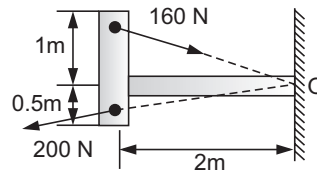


Fig. Prob. 4

5. Resolve 300 N force in to components along u and v directions. Refer *figure Prob.5*.

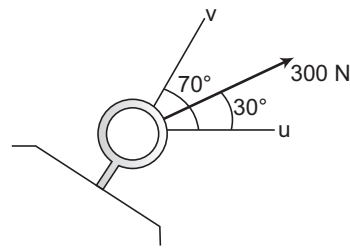


Fig. Prob. 5

6. Resolve 300 N force in to components along u and v directions. Refer *figure Prob.6*.

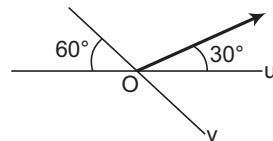


Fig. Prob. 6

7. Two forces of 150 N and 110 N are acting on a hook as shown in *figure Prob.7*. Determine (a) magnitude and direction of resultant of these two forces. (b) The magnitudes of two other forces P_u and P_v that would have the same resultant.

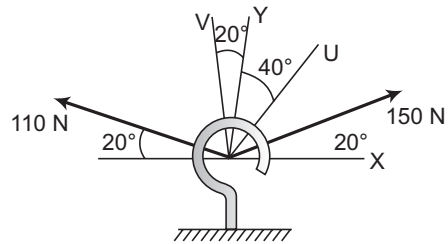


Fig. Prob. 7

8. Determine magnitude of the resultant force and its direction from x -axis of three concurrent forces as shown in *figure Prob.8*.

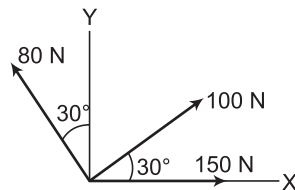


Fig. Prob. 8

9. A rectangular plate is subjected to three concurrent forces as shown in *figure Prob.9*. Determine magnitude of the resultant force and its direction from x -axis.

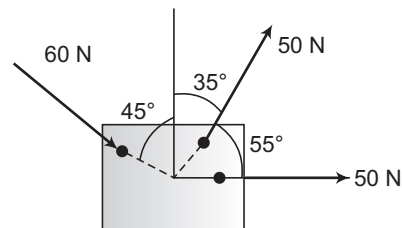


Fig. Prob. 9

10. Determine magnitude of the resultant force and its direction from x -axis of three concurrent forces as shown in *figure Prob.10*.

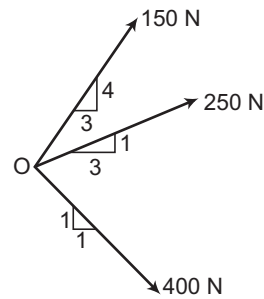


Fig. Prob. 10

11. Determine magnitude of the resultant force and its direction from x -axis of four concurrent forces as shown in *figure Prob.11*.

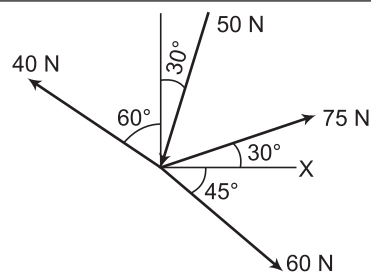


Fig. Prob. 11

12. The magnitude of the resultant of three concurrent forces is 45 N and directed along positive y axis. Determine magnitude of force P . Refer *figure Prob.12*

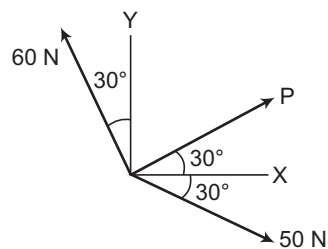


Fig. Prob. 12

13. Determine the moments of the 120 N force shown in *figure Prob.13* about points A , B and C .

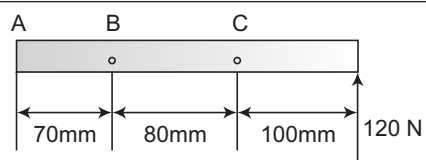


Fig. Prob. 13

14. Determine moment of each forces shown in *figure Prob.14* about points O .

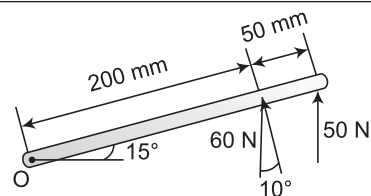


Fig. Prob. 14

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15. Two forces are applied to a bracket as shown in *figure Prob.15*. Determine the sum of the moments of the two forces about point *A* and point *B*.

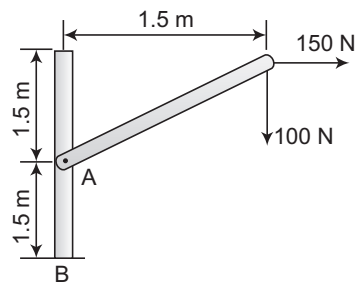


Fig. Prob. 15

16. Three forces are applied to a bracket as shown in *figure Prob.16*. Determine the sum of the moments of the three forces about point *O*.

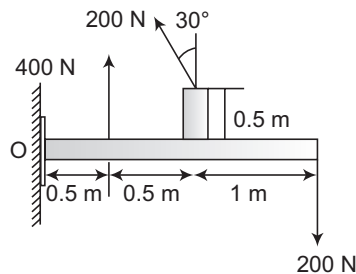


Fig. Prob. 16

17. The moment exerted by the weight *W* about point *B* is 300 N.m. Determine the moment by the weight *W* about point *A*. Refer *figure Prob.17*.

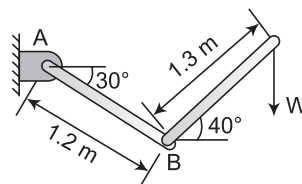


Fig. Prob. 17

18. Two forces are applied to a circular plate as shown in *figure Prob.18*. determine moment of each forces about point *O*.

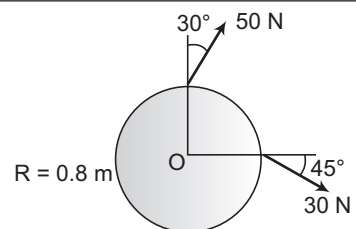


Fig. Prob. 18

19. Determine the resultant of the three forces and the couple *C* acting on a rectangular plate as shown in *figure Prob.19*.

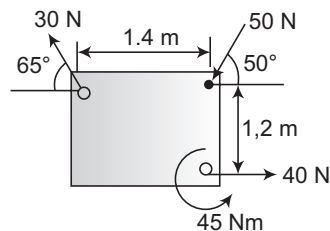


Fig. Prob. 19

20. Determine the resultant of the three forces acting on a beam as shown in *figure Prob.20*.

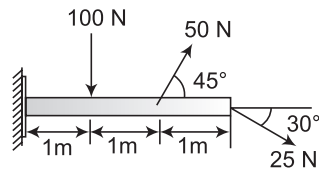


Fig. Prob. 20

21. Three parallel forces are acting on a beam as shown in *figure Prob.21*. Determine the resultant force and its location with respect to support A.

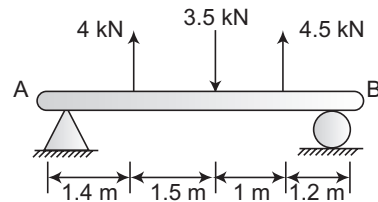


Fig. Prob. 21

22. The resultant of the force system shown in *figure Prob.22* is a 120 N.mm anticlockwise couple. Determine magnitude of P_1 , P_2 and C.

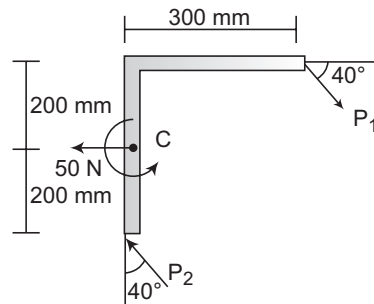


Fig. Prob. 22

23. Determine the moment of the couple shown in *figure Prob.23*. Also determine perpendicular distance between the two forces.

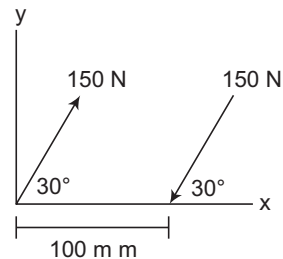


Fig. Prob. 23

24. If $P = 80 \text{ N}$ determine the couple moment produced. Refer *figure Prob.24*.

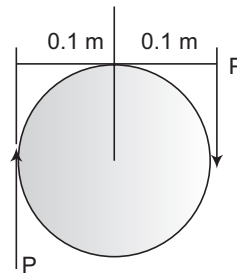


Fig. Prob. 24

25. In Prob. No.24, If the couple moment produced by two forces P is 30 N.m . Determine magnitude of P .

26. A rectangular plate is subjected to two forces and the couple as shown in *figure Prob. 26*. Replace the force system with an equivalent force – couple system at point O

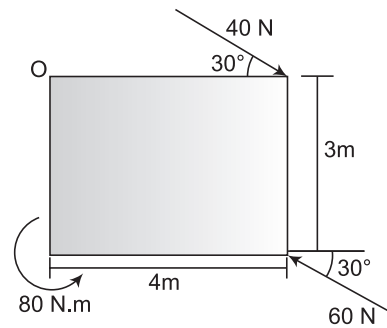


Fig. Prob. 26

27. A bracket is subjected to three forces as shown in *figure Prob. 27*. Replace the force system with an equivalent force – couple system at point A .

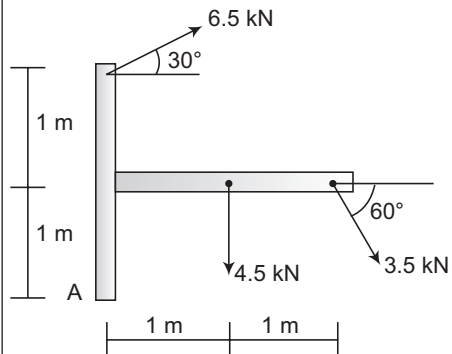


Fig. Prob. 27